

Program overview

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Year 2024/2025
Organization Aerospace Engineering
Education Bachelor Aerospace Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
LR Bachelor 2024		AE Bachelor 2024					
LR BSc 1e jaar 2024		AE BSc 1st year 2024					
AE1108	Aerospace Materials and Structures	6					
AE1108-I	Aerospace Materials	3					
AE1108-II	Aerospace Mechanics of Materials	3					
AE1110	Introduction to Aerospace Engineering	9					
AE1110-I	Introduction to Aerospace Engineering I	5					
AE1110-II	Introduction to Aerospace Engineering II	4					
AE1111	Exploring Aerospace Engineering & Design	5					
AE1111-I	Exploring Aerospace Engineering	3					
AE1111-II	Engineering Drawing	2					
AE1111-III	Mentor Programme	0					
AE1130	Engineering Mechanics	7					
AE1130-I	Statics	4					
AE1130-II	Dynamics	3					
AE1205	Programming & Scientific Computing in Python for AE1	2					
AE1222	Aerospace Design & Construction	9					
AE1222-I	Design & Construction	5					
AE1222-II	Aerospace Design & Systems Engineering Elements	4					
AE1241	Physics	6					
WI1402LR	Calculus II	5					
WI1403LR	Linear Algebra	5					
WI1421LR	Calculus I	6					
LR BSc 2e jaar 2024		AE BSc 2nd year 2024					
AE2111	Aerospace System Design	8					
AE2111-I	Systems Design	5					
AE2111-II	Aerospace Design & Systems Engineering Elements	3					
AE2130	Aerodynamics Sub- and Supersonic	7					
AE2130-I	Aerodynamics I	3					
AE2130-II	Low Speed Wind Tunnel Test	1					
AE2130-III	Aerodynamics II	3					
AE2135	Structural and Vibrational Analysis & Design	8					
AE2135-I	Structural Analysis & Design	5					
AE2135-II	Vibrations	3					
AE2220	Applied Numerical Analysis and Computational Modelling	6					
AE2220-I	Applied Numerical Analysis	3					
AE2220-II	Computational Modelling	3					
AE2224	Test, Analysis and Simulation	8					
AE2224-I	Test, Analysis & Simulation	5					
AE2224-II	Artificial Intelligence for Aerospace Engineering	3					
AE2230	Flight & Orbital Mechanics and Propulsion	8					
AE2230-I	Flight & Orbital Mechanics	4					
AE2230-II	Propulsion & Power	4					
AE2235	Aerospace Signals, Systems & Control	7					
AE2235-I	Aerospace Systems & Control Theory	4					
AE2235-II	Signal Analysis and Telecommunication Principles	3					
WI2180LR	Differential Equations & Probability and Statistics	8					
WI2180LR-I	Differential equations	4					
WI2180LR-II	Probability and Statistics	4					
LR BSc 3e jaar 2024		AE BSc 3rd year 2024					
LR BSc 3e jaar Major		AE BSc 3rd Year Major					
AE3200	Design Synthesis	15					
AE3211	Aerospace Systems Engineering, Design & Production	6					
AE3211-I	Systems Engineering & Aerospace Design	3					
AE3211-II	Production of Aerospace Systems	3					
AE3212	Aerospace Flight Dynamics, SVV including Flight Test	9					
AE3212-I	Aerospace Flight Dynamics & Simulation, including Flight Test	5					
AE3212-II	Simulation, Verification & Validation	4					
LR BSc 3e jaar Minors		AE BSc 3rd Year Minors					

Year	2024/2025
Organization	Aerospace Engineering
Education	Bachelor Aerospace Engineering

LR Bachelor 2024

Year	2024/2025
Organization	Aerospace Engineering
Education	Bachelor Aerospace Engineering

LR BSc 1e jaar 2024

AE1108	Aerospace Materials and Structures	6
Contact Hours / Week x/x/x/x	N/A	
Education Period	2 3	
Start Education	2	
Exam Period	2 3 4	
Course Language	English	
Course Contents	This module deals with the fundamentals of Material science and of Mechanics of Materials. Aerospace Materials is a course that teaches the fundamentals of material science needed for an understanding of the behaviour of materials in a structure. Aerospace Mechanics of Materials is, just like the Statics and Dynamics courses, part of the foundation of the mechanics on which much of aerospace engineering is based. The module is taught in two separate courses: Aerospace Materials and Aerospace Mechanics of Materials. Please see the course description of the underlying courses for more details regarding the contents and set up of the course.	
Study Goals	At the end of the module students will be able to independently solve problems related to Aerospace materials and Mechanics of materials. For more detailed learning objectives please see the learning objectives of the underlying courses.	
Education Method	The general method of teaching for this module is a combination of lectures and (for the mechanics of materials) practice sessions. Students are expected to take active part in class and during the instruction sessions as well as practice a lot at home. The old adagio: "Practice makes Perfect" definitely applies to this module and students are urged to plan accordingly. For more details please see course descriptions.	
Assessment	Each course will be assessed separately. See the course descriptions for more details. The final grade of the module AE1108 - Aerospace Materials and Structures (rounded to the nearest half) will be determined by the following formula: $\text{grade AE1108} = \{(\text{grade AE1108-I}) * 3 + (\text{grade AE1108-II}) * 3\} / 6$ Each course grade (registered to one decimal place) must be 5.0 or higher AND the overall grade (rounded to the nearest half) for AE1108 must be 6.0 or higher in order to pass this module. No partial credits will be awarded for the individual courses of this module.	

AE1108-I	Aerospace Materials	3
Responsible Instructor	Dr. S.J. Garcia Espallargas	
Instructor	Dr. B. Caglar	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This course deals with the understanding of material properties and performance and how these are connected to basic structural behaviour. This includes learning the reasons behind certain material properties and how to change some of them. Furthermore, students will learn concepts as stresses and strain in tensile, shear or bending and how to apply these in basic structural elements. Thermal properties and their origin will also be addressed. As durability of aerospace structures is crucial, also the corrosion resistance and other degradation considerations will be discussed. Finally, the link between material properties and material selection for design considering the environmental impact of materials production and use is made. The course will cover the following aspects in an aerospace materials context: -Material-related concepts such as elastic modulus, stiffness, plasticity, yield, ductility, strength, toughness, hardening, Tg, Tm, thermoset, thermoplastic, elastomer, electrochemical potential, electrical/thermal conductivity -Microstructure-dependent and microstructure-independent properties -Material properties and their relation to material composition and microstructure -Development and use of material indices for material selection in a given design -Stiffness-limited design -Strength-limited design -Fracture-limited design -Fatigue-limited design -Materials behavior at high temperatures -Materials degradation (corrosion, oxidation) -Environmental considerations of the impact of material choice and use (circular economy) -Strategic thinking to select materials for a target design	
Study Goals	At the end of this course, the student will be able to: -Implement strategic thinking for material selection; -Link the material properties to well-functioning simplified aerospace components; -Describe and discuss the relationship between the composition and structure of a material and its properties; -Describe the fundamental reasons behind material behaviour (mechanical, thermal, corrosion, etc.); -Derive and use material indices and material charts to select materials for a given simple geometry and loading conditions; -Select materials considering application, environmental resistance, and sustainability and ethical aspects; -Design simple products considering material properties and loading conditions.	
Education Method	Lectures	
Literature and Study Materials	-Book chapters covered during the course from the book: "Materials: Engineering, Science, processing and design", M. Ashby, H. Shercliff and D. Cebon. 4th edition. Elsevier, ISBN 9780081023778 or 9780081023761. The 3rd edition may also be used, but the students need to be aware of the differences with the 4th edition. The slides and lectures are based on the 4th edition -Lecture slides/recordings available on Brightspace -Content of the weekly question-based lectures -Support videos available on Brightspace -Weekly recommended exercises -Provided exam examples	
Books	"Materials: Engineering, Science, processing and design", M. Ashby, H. Shercliff and D. Cebon. 4th edition. Elsevier, ISBN 9780081023778 or 9780081023761. The 3rd edition may also be used, but the students need to be aware of the differences with the 4th edition. The slides and lectures are based on the 4th edition.	
Assessment	Written exam. (closed book)	
Set-up	The teaching method follows a "flipped classroom" style to maximize learning and grow in independence. This method implies the students study at the beginning of the week and deepen or/and clarify knowledge during the weekly question-based lectures and group work contact hours at the end of the week. The course is structured in 6+1 weeks (Week 7 will only consist of one lecture of two hours, Lecture 13, dedicated to exam-like questions and a material selection software). This education method implies the students MUST visualize the weekly videos available on Brightspace OR/AND read the related study material from the book AND post/send their questions BEFORE the question-based lecture programmed each week. Students are expected to check the Brightspace page on a regular basis (i.e. weekly). Each week consists of the following steps: (1) Monday to Wednesday. Students work on/study the "weekly assigned workload" available on the course Brightspace page and post related questions. As a student you can plan around 4-5h/week for this pre-work. The "weekly assigned workload" is facilitated as pre-recorded videos and related pdf of the slides on Brightspace. Most content is also available in the course book for self-reading. The lecture videos (and pdf) available on Brightspace are grouped per week, with each week containing two lectures (e.g. Week 1= lecture 1 + lecture 2, week 2=lecture 3 + lecture 4, etc). Until Wednesday evening the students are expected to "send questions" using the 'collaboration' tab on Brightspace or e-mail to TAs to remain anonymous. The questions can be about things that are unclear for that week (e.g. I did not understand well the concept of stress concentration factor) or about deepening in the course content (e.g. what is the implication of stress concentration in design). This is a very important step as the questions are the starting point for the staff to build the weekly "question-based lecture". (2) Thursday. The first hour is planned as the "Question-based lecture" as allocated in the study plan. This is a lecture based on the questions submitted by the students for that week. During the second hour the students will have the opportunity to attend research seminars given by PhD candidates working on aerospace materials science & engineering topics.	

(3) Friday. The first hour the students will work in predefined groups on the recommended exercises for that week. There will be one TA per room to assist and solve questions.
The second hour will be dedicated to the so-called Discovery Question. The student groups from the first hour will work on a pre-assigned research topic under the supervision of a TA. The research topics are related to current material science and engineering topics of potential interest for aerospace and space applications. More details will be given during the first lecture.

AE1108-II	<i>Aerospace Mechanics of Materials</i>	3
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Instructor	Dr. N. Yue	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	AE1130-I: Statics - Specifically, students are expected to be comfortable in drawing free body diagrams, solving for internal forces and moments, determining cross sectional properties such as 1st and 2nd moment of area, and determining/plotting internal force and moment diagrams.	
Course Contents	This course deals with understanding of basic structural behaviour. Students will learn about concepts such as stress and strain under different kinds of loading (e.g. tensile, shear and bending). Additionally, students will develop skills to solve statically indeterminate structural problems.	
Study Goals	At the end of this course, the student will be able to: Use mechanics to solve basic problems dealing with the stress/strain/displacement of structures in static equilibrium under mechanical & thermal loads. Use mechanics to solve basic problems dealing with the relation between material behaviour & structural behaviour under mechanical & thermal loads.	
Education Method	Two lectures and 1 instruction per week with weekly quizzes to be completed each week. Preparation for lectures is mandatory with assigned readings/videos to be completed before each lecture.	
Literature and Study Materials	Hibbeler Mechanics of Materials 10th edition SI Units ISBN-13: 978-1-292-17820-2	
Books	Mechanics of Materials, by R.C. Hibbeler, SI 10th edition	
Assessment	Weekly quizzes that will be given indicative grades for feedback, but do not count towards the final grade (formative feedback). A list of recommended problems will be provided that can be used to practice for the weekly quizzes. Written final exam (with an entry requirement) covering all course learning objectives worth 10 points (100% of final grade). A resit exam opportunity (with an entry requirement) will be offered in the exam period of the subsequent quarter in which the course is offered. The resit will be worth 10 points (100% of final grade). Entry requirement for the final/resit exam: Students are eligible to take the final exam within the academic year in which a student has met the threshold for the number of required honest attempts for weekly quizzes. An attempt at a particular weekly quiz is seen as honest, if the paperwork that was submitted shows that a clear attempt was made to solve the problem(s) given in the particular weekly quiz. Students who did not meet the entry requirement for the final exam will be offered a repair opportunity for the resit exam. More details about the weekly quizzes and requirements for entry in the final exam and repair thereof will be explained in the first lecture of the course.	
Set-up	For more information see the AE1108-II course information on Brightspace. Students prepare for lectures by watching a series of theory-based videos. Students then attend the active lectures, in which students apply the theory learned through the videos to solve relevant engineering problems. After the active lecture, students undertake self-study on practice problems and weekly quizzes to be completed each week. Once a week they will attend an instruction in which they practice problems from the book under supervision of a teaching-assistant.	

AE1110	Introduction to Aerospace Engineering	9
Module Manager	Prof.dr.ir. J.M. Hoekstra	
Contact Hours / Week x/x/x/x	see AE1110-I and AE1110-II	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course will introduce the broadness of Aerospace Engineering and explain its fundamentals.	
Study Goals	Basic understanding and overview of field of Aerospace Engineering.	
Education Method	Lectures, some classroom sessions and materials& structures practical.	
Assessment	Each course will be assessed separately. See the course descriptions for more details. The final grade of the module AE1110 - Introduction to Aerospace Engineering (rounded to the nearest half) will be determined by the following formula: $\text{grade AE1110} = \{(\text{grade AE1110-I}) * 5 + (\text{grade AE1110-II}) * 4\} / 9$ Each course grade (registered to one decimal place) must be 5.0 or higher AND the overall grade (rounded to the nearest half) for AE1110 must be 6.0 or higher in order to pass this module. No partial credits will be awarded for the individual courses of this module.	

AE1110-I	Introduction to Aerospace Engineering I	5
Responsible Instructor	Prof.dr.ir. J.M. Hoekstra	
Instructor	Ir. J.A. Melkert	
Instructor	Prof.dr. F. Scarano	
Instructor	C.D. Rans	
Contact Hours / Week x/x/x/x	6/0/0/0 (instruction) + 2/0/0/0 studio class room	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This first part of the course Introduction to Aerospace Engineering presents an overall picture of the aeronautics domain. This overview involves a number of different perspectives on the aerospace domain, and shows some basic principles of the most important concepts for flight. Then a part on Flight Performance is given, with the equations of motion and endurance, range, envelope, etc. Finally an introduction is given to the underlying aerodynamics, both incompressible flow (subsonic) as well as compressible flow(trans- and supersonic).	
Study Goals	General knowledge of aircraft and their systems (such as avionics & propulsion) and its history Derive & apply atmospheric calculations Know and apply forces on an aircraft: Lift, drag, thrust. Derive power & efficiency equations. Derive & Apply equations for moments, stability & apply on aircraft configuration Structure & Materials: understand stress and basic structural elements of an aircraft Apply control volume approaches Explain flow regimes (compressible/incompressible aerodynamics) and to estimate viscous and thermal effects Derive & apply equations of motion to determine aircraft performance in steady gliding, horizontal and climbing flight Derive aircraft performance diagram and flight envelope, in relation to aircraft morphology, lift-drag polar and engine performance	
Education Method	Lectures (live streamed, links on BrightSpace)	
Literature and Study Materials	Introduction to Flight, by John D. Anderson; 6th edition or later; ISBN of 7th edition: 9780071086059 Handouts	
Assessment	Exam consists of three different tests. Each of the tests is an on-site exam. The format of the tests is written (parts 1 & 3) and digital (part 2) in three separately scheduled sessions. The three different parts of Intro to AE I will together form one grade. There is no minimum score for each of the parts. The average accounts for the exam grade. There is one session for a Resit in January. In this session each part can be redone separately. The recalculated grade is then the Resit grade. But be aware that the partial scores are only kept within the academic year, as after January only a course grade will be kept.	
Set-up	Together with the course "Introduction to Aerospace Engineering II" (AE1110-II) this course is strongly related to the Project "Exploring Aerospace Engineering" (AE1111-I) which is running parallel in the first semester.	

AE1110-II	Introduction to Aerospace Engineering II	4
Responsible Instructor	Prof.dr.ing. R.C. Alderliesten	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	This part of the course Introduction to Aerospace Engineering is focused on two aerospace disciplines: "spaceflight" and "structures and materials". These topics are discussed in detail and will provide an understanding for both aircraft and space missions.	
Study Goals	- List/describe the reasons for going into space and the principles of rockets, including their trajectories. - Motivate the selection of spacecraft configurations depending on the mission and identify the main elements of a satellite. - Describe the features of the space environment and their consequences for space activities. - Determine elementary satellite orbits, transfer orbits and maneuvers. - Describe and work with elementary space propulsion aspects: launch, velocity budget and rocket equation. - List the characteristics of typical aerospace materials & structures and describe their meaning and relevance. - Identify the main structural elements and components of an aerospace vehicle and describe their functions and performance. - Perform elementary calculations to determine structural strength, stiffness, and fatigue properties	
Education Method	Lecture	
Literature and Study Materials	Introduction to Flight, by John D. Anderson; 6th edition; ISBN-13: 978-007-126318-4. Spacecraft Systems Engineering, by Fortescue, Stark and Swinerd; 3rd. rev. edition; ISBN-13: 978-0471619512 (recommended reading only). Powerpoint presentations. Introduction to Aerospace Structures and Materials, by Rene Alderliesten (https://textbooks.open.tudelft.nl/index.php/textbooks/catalog/15). Lecture slides and videos.	
Assessment	Multiple choice exam	
Set-up	Together with the course "Introduction to Aerospace Engineering I" (AE1110-I) this course is strongly related to the Project "Exploring Aerospace Engineering" (AE1111-I) which is running parallel in the first semester.	

AE1111	Exploring Aerospace Engineering & Design	5
Module Manager	Ir. P.C. Roling	
Contact Hours / Week x/x/x/x	see AE1111-I, AE1111-II and AE1111-II	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This module consists of two courses: AE1111-I Exploring Aerospace Engineering AE1111-II Engineering Drawing Both courses must be completed to get the 5 credits in the module.	
Study Goals	This module aims to apply the basic skills they will require to study aerospace Engineering: - Learn to study at a university level. - Communicate by sketching and drawing your designs. - Explore the basic principles of Aerospace Engineering.	
Education Method	AE1111-I: Once (Q2) to twice (Q1) a week a four hour project session AE1111-II: Once a week a four hour session in an instruction room / computer lab.	
Assessment	Each course will be assessed separately. See the course descriptions for more details. The final grade of the module AE1111 - Exploring Aerospace Engineering & Design(rounded to the nearest half) will be determined by the following formula: $\text{grade AE1111} = \{(\text{grade AE1111-I}) \cdot 3 + (\text{grade AE1111-II}) \cdot 2\} / 5$ Each course grade (registered to one decimal place) must be 5.0 or higher AND AE1111-III must have a pass grade (V) AND the overall grade (rounded to the nearest half) for AE1111 must be 6.0 or higher in order to pass this module. No partial credits will be awarded for the individual courses of this module.	

AE1111-I	Exploring Aerospace Engineering	3
Responsible Instructor	Ir. P.C. Roling	
Instructor	Ing. G.P.A. Wijnschenk Dom	
Contact Hours / Week x/x/x/x	8/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	In your project groups of approximately 10 people you will be analysing the design of a flying wing for a mission on Mars. During the project you will be asked to perform assignments related to the design of this wing including small experiments in aerodynamics, flight and orbital mechanics. You will also design, build and fly a model of a flying wing. You will develop your team working skills and reflect on them. You will be required to report your results after each session in a small report and summarize the outcomes of your work in a scientific poster. This project is related to the Introduction to Aerospace Engineering courses (AE1110).	
Study Goals	At the end of the project you must be able to: - explore the reasons behind the design of an aerospace vehicle - apply the knowledge from the related courses - work in small groups and reflect on your role within the group	
Education Method	Thematic group project	
Computer Use	As reports will need to be produced, laptops are strongly recommended.	
Literature and Study Materials	Project guide AE1111-I Exploring Aerospace Engineering Brightspace	
Assessment	The grade consists of the following partial grades: Group mark scientific poster (20%) Individual mark evaluation group skills (40%) An individual E-test (40%) Grades will be determined per the requirements listed below: 1. The whole project has been completed 2. All rules regarding absence are met 3. All compensatory assignments have been completed successfully (if applicable) 4. No more than one partial grade is lower than 6.0 5. All partial grades are 5.0 or higher If any of the conditions listed under 1-5 are not met, a final grade of 1.0 is awarded for the project. If a grade of 1.0 is awarded to a student, the entire project must be redone the following academic year. Both partial and final grades will be rounded to 1 decimal. The following rules regarding absence hold: 1. Students are allowed to miss a maximum of an average week worth of sessions per period, equating to eight hours for the first period and four hours for the second period. 2. If a project has a supporting course the student is only allowed to be absent for one session of this course. This session is not counted as a missed project session as mentioned under 1. 3. Missing time (either by being late or leaving at any time) during project sessions can result in being registered as missed sessions. 4. Missed sessions must be compensated according to the rules stated below. 5. The student must attend the first week of a project. Not being present in the first week of the project results in exclusion from participation in the project in that academic year. 6. Absence for whatever reason for more than the allowed number of sessions or failure to make up for missed hours will mean that the student cannot obtain a pass grade for the project. 7. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic Counsellors for guidance. For compensating absence, the following applies: 1. The student is responsible for fulfilling the compensatory assignment, at a time or within a time limit set by the responsible lecturer (this also applies to supporting courses). 2. The quality of the compensatory assignment is assessed by the responsible lecturer. If the student does not complete the assignment within the allocated time and/or with sufficient quality, they will fail the project. 3. Compensatory assignments cannot be amended or redone if deemed unsatisfactory upon assessment by the responsible lecturer and results in failing the project. 4. No opportunities are offered outside the duration of the project for making up for missed sessions. 5. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic Counsellors for guidance.	

AE1111-II	Engineering Drawing	2
Responsible Instructor	E.J. van den Bos	
Instructor	Ing. G.P.A. Wijnschen Dom	
Responsible for assignments	D.M. Atherstone	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1 2	
Exam Period	none	
Course Language	English	
Parts	The course has three main components: 1. Basic Sketching. 2. Common Conventions for Technical Drawings. 3. 3D CAD using the 3DEXPERIENCE CATIA software.	
Course Contents	You will learn skills to register the geometry of engineering solutions. You will train professional sketching techniques by doing assignments. You will get acquainted with the conventions for technical drawings and you will make computer based 3D models and drawings of Aerospace related parts and assemblies. The skills you learn will be used in the course of your education in projects and other assignments where the geometry of a design needs to be documented.	
Study Goals	At the end of this course, the student will: 1 Be able to sketch straight lines and fluent circles and ellipses. 2 Be able to use the ellipse rule for correct placement of circular and cylindrical items. 3 Be able to use 1-, 2- and 3-point perspective in their work. 4 Be able to adopt and use perspective from a given starting point (photo or CAD print). 5 Be able to shade cubical items correctly. 6 Be able to shade cylindrical items correctly. 7 Be able to use markers for efficient colored backgrounds. 8 Be able to use communicative additions, such as arrows, text and bullets to enhance the communicative value of a sketch. 9 Be able to recognize and apply, internationally agreed upon drawing conventions. 10 Be able to read and interpret engineering drawings. 11 Be able to make Detail and Assembly drawings, compliant to the taught drawing conventions. Be able to use the 3DEXPERIENCE software for: 12 Making feature based parametric surface and solid models. 13 Making assemblies. 14 Collaboration on a group assembly. 15 Designing a multi-part assembly in a group context.	
Education Method	For Basic Sketching, the Sketchdrive® platform will be made available to you via Brightspace. Sketchdrive® contains instruction videos for you to be able to perform the assignments in the Basic Sketching Workbook. Results have to be uploaded to your Sketchdrive® portfolio. You will have two practical sessions per week, each session lasts two hours. At the start of each session a video will be made available with demos and feedback. The course component Common Conventions for Technical Drawings is taught through guided self-study. A quiz will be made available on Brightspace every week of the first quarter. A summative computer test is scheduled after the first quarter. You must pass six out of seven weekly quizzes. The quiz topics match the available videos on Brightspace, so by studying the video you will gain the knowledge to complete the quiz successfully. The knowledge is required to pass the test, but also to be able to do the first CAD assessments. A lecture is scheduled at the end of the first quarter to prepare you for the test. For the CAD part of the course you will also have two practical sessions per week, each session lasting two hours. It requires a Windows-based installation of the 3DEXPERIENCE software. A license for the software will be made available to you during the course. The CAD assignments will be accompanied by an online dashboard and interactive tutorials. Staff and TA's will be available for answering questions and feedback at all practical sessions. Please be aware of the fact that the 3DEXPERIENCE software does not run under macOS.	
Computer Use	The 3DEXPERIENCE software is a Windows based platform, an up-to-date laptop is required to be able to install the software. We strongly recommend to make use of the TU Delft laptop project possibility.	
Books	Basic Sketching Workbook Common Conventions for Technical Drawings	
Reader	Basic Sketching for Aerospace Engineering	
Assessment	Assessments: During the course the student will be asked to upload a selection of the work to the Sketchdrive® platform each week. This will allow the sketching staff to give the student feedback and set a grade as the student progresses. Basic Sketching will be graded based on the indicated Workbook Sketchdrive® assignments. The assessment criteria are: - Presence of straight lines and fluent circles and ellipses - Application of the ellipse rule - Application of perspective (general) - Ability of adopting perspective from a given starting point - Correct shading of cubical items - Correct shading of cylindrical items - Application of efficient colored backgrounds - Use of arrows, text and bullets for storytelling Common Conventions for Technical Drawings will be graded based on the result of the summative computer test assignment.	

	The CAD grade is determined by the different subgrades: Assignment 8 Detail Drawing. Assignment 9 Assembly Drawing. Assignment 12 3D Modelling and Collaboration. Final Synthesis Assignment.
Remarks	For the sketching sessions some drawing materials are required and mandatory. Details on availability and sales of these materials will be published on the Brightspace page of this course.

AE1111-III	Mentor Programme	0
Responsible Instructor	Dr.ir. D.M.J. Peeters	
Contact Hours / Week x/x/x/x	x/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This mentor programme is to support, stimulate and motivate first year students in starting university. All first year students are assigned to a mentor group where different aspects of studying at Aerospace Engineering are addressed.	
Study Goals	n/a	
Education Method	n/a	
Assessment	n/a	

AE1130	Engineering Mechanics	7
Module Manager	C.D. Rans	
Contact Hours / Week x/x/x/x	8/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Expected prior knowledge	High School Physics and Maths	
Course Contents	This module deals with the fundamentals of Engineering Mechanics: Statics & Dynamics. It is a natural continuation from Newtonian Mechanics and laws of conservation of energy from high school. This module forms the foundation of the mechanics on which much of aerospace engineering is based. The module is taught in two separate courses: Statics and Dynamics. Please see the course description of the underlying courses for more details regarding the contents and set up of the course.	
Study Goals	At the end of the module students will be able to independently solve problems related to Statics and Dynamics by selecting the appropriate solution method. For more detailed learning objectives please see the learning objectives of the underlying courses.	
Education Method	The general method of teaching and assessment for this module is a combination of lectures with interactive practice sessions predominantly using computer based exercises and tests. Students are expected to take active part in class and during the instruction & studio classroom sessions as well as practice a lot at home. The old adagio: "Practice makes Perfect" definitely applies to this module and students are urged to plan accordingly. For more details please see course descriptions	
Assessment	Each course will be assessed separately. See the course descriptions for more details. The final grade of the module AE1130 - Engineering Mechanics (rounded to the nearest half) will be determined by the following formula: $\text{grade AE1130} = \{(\text{grade AE1130-I}) * 4 + (\text{grade AE1130-II}) * 3\} / 7$ Each course grade (registered to one decimal place) must be 5.0 or higher AND the overall grade (rounded to the nearest half) for AE1130 must be 6.0 or higher in order to pass this module. No partial credits will be awarded for the individual courses of this module.	

AE1130-I	Statics	4
Responsible Instructor	C.D. Rans	
Instructor	Dr. J.J.E. Teuwen	
Contact Hours / Week x/x/x/x	Lecture 6/0/0/0, instructions 2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	High School Physics & Mathematics	
Parts	1. Forces and Equilibrium of a Particle: decomposition and sum of forces using both an analytical approach (vector algebra) and a graphical approach (e.g. force polygon); Static equilibrium of a particle in 2D and 3D. (Ch. 1, 2, and 3) 2. Equilibrium of Rigid Bodies: moment of a force; equivalent force systems; free body diagrams; supports and connections (e.g. hinges); reaction forces and moments; 2D equilibrium of rigid bodies. (Ch. 4 and 5) 3. Structural Analysis: calculation of forces in trusses using method of joints and method of sections; analysis of frames; analysis of machines. (Ch. 6) 4. Internal Forces: evaluation of normal force, shear force, bending moment and torsional moment in a bar; notation and sign conventions; differential equations; relations between normal force, shear force and bending moment diagrams. (Ch. 7) 5. 3D extension of methods: equilibrium in 3D; analysis of trusses, frames, and machines in 3D; internal force diagrams in 3D. (Ch. 5.5, 5.6, 5.7, 6.5, 7) 6. Modelling Approximations: representing a body by its centre of mass; representing a body by its centre of gravity; representing a beam cross-section by its moment of inertia; representing a body by its mass moment of inertia; parallel axis theorem. (Ch. 9.1-9.3, 10.1, 10.2, 10.4, 10.8) 7. Virtual Work: principle of virtual work as an alternative formulation for the equilibrium of mechanical systems; application to the calculation of reaction forces and moments and internal forces and moments. (Chapter: 11.1 - 11.3)	
Course Contents	Statics deals with the principles of equilibrium. In this course, the principles of forces and moments will be explained as well as principle of equilibrium of forces and moments. This also includes the equilibrium of 2D and 3D structures and trusses. Furthermore the principle of internal forces and moments is addressed as well as the use of the principle of virtual work to calculate both external and internal loads. Finally, the concepts of centre of gravity, centroids and moments of inertia are discussed.	
Study Goals	At the end of this course, the student will be able to solve basic problems dealing with statics. They will be able to: - Define the basic terms in statics such as force, moment, body using the appropriate units and notations - Use Newtons 1st law to determine the reactions on 2D and 3D bodies and particles in equilibrium - Calculate the centroids and mass and area moments of inertia of elementary shapes (including thin-walled structures) - Calculate the normal forces in a truss structure - Calculate and construct diagrams of the distribution of internal normal forces, shear forces, bending moments and torsion moments as a result of external loading including distributed loads following a standard definition and recognise when diagrams do not follow this definition - Use the principle of virtual work to derive internal and external forces and moments of structures and systems	
Education Method	Online content, lectures and instructions	
Literature and Study Materials	Engineering Mechanics - Statics by R.C. Hibbeler, 15th edition in SI units, Pearson Prentice Hall incl. Mastering Engineering access code: ISBN 9781292444048 VSV Leonardo da Vinci offers the Mechanics package with a special discount and three-year access to Mastering Engineering. Mastering Engineering is an online portal for the textbook that includes additional video solutions and study resources.	
Assessment	Weekly quizzes that will be given indicative grades for feedback, but do not count towards the final grade (formative feedback). A list of recommended problems will be provided that can be used to practice for the weekly quizzes. Written final exam (with an entry requirement) covering all course learning objectives worth 10 points (100% of final grade). A resit exam opportunity (with an entry requirement) will be offered in the exam period of the subsequent quarter in which the course is offered. The resit will be worth 10 points (100% of final grade). Entry requirement for the final/resit exam: Students are eligible to take the final exam within the academic year in which a student has met the threshold for the number of required honest attempts for weekly quizzes. An attempt at a particular weekly quiz is seen as honest, if the paperwork that was submitted shows that a clear attempt was made to solve the problem(s) given in the particular weekly quiz. Students who did not meet the entry requirement for the final exam will be offered a repair opportunity for the resit exam. More details about the weekly quizzes and requirements for entry in the final exam and repair thereof will be explained in the first lecture of the course. All course material (book, lecture slides, assignments) will be in English. To avoid confusion between English and Dutch language terminology all relevant terms will be taught in English. Therefore all exams will be in English only. This includes the number format used (decimal point instead of comma, etc.).	
Permitted Materials during Tests	Closed book exam. Necessary formulas will be provided during the exam. For the quizzes and final exam, only the use of the standard bachelor calculator (Texas Instruments (TI) 30xb multiview) is allowed. Writing on the exam must be in pen.	
Set-up	Students are expected to study the online lecture preparation content prior to the lecture. Students attend the active lecture, in which new topics are introduced and examples are completed with student input to provide insight. After the active lecture, students undertake self-study either together or alone, this can be based on the list of recommended problems.	

Once a week they are expected to attend an instruction in which they practice problems from the book under supervision of a teaching-assistant.

Each week, there will be a quiz (exam level and in a practice exam-setting) which will be graded. These quizzes are offered once, and have no resit opportunity.

AE1130-II	Dynamics	3
Responsible Instructor	Dr.ing. S. Giovanni Pereira Castro	
Instructor	X. Wang	
Contact Hours / Week x/x/x/x	Lecture 0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Parts	Topics: 1. Kinematics of a particle (chapter 12). 2. Kinetics of a particle: Force and acceleration (chapter 13). 3. Kinetics of a particle: Work and energy (chapter 14). 4. Kinetics of a particle: Impulse and momentum (chapter 15). 5. Planar kinematics of a rigid body (chapter 16). 6. Planar kinetics of a rigid body: Force and acceleration (chapter 17). 7. Planar kinetics of a rigid body: Work and energy (chapter 18). 8. Planar kinetics of a rigid body: Impulse and momentum (chapter 19).	
Course Contents	This course introduces the basic concepts of kinematics and kinetics that are necessary to understand, analyze and design a variety of mechanisms and machines, such as airplanes, automobiles or satellites. Dynamics is concerned with the description and analysis of the movement of particles and rigid bodies (kinematics) and the relations between kinematics and kinetics (mass and forces). The course covers the equations of motion and derived methods of analysis, namely the work-energy relation and the linear and angular impulse-momentum relations.	
Study Goals	The goal is to provide the necessary knowledge, insight and skills to understand and quantify the motion of particles and rigid bodies (kinematics) and the relation between forces and motion (kinetics). At the end of this course, the student should be able to: 1. Calculate motion of a particle using three different coordinate systems (rectangular, n-t and cylindrical) 2. Calculate dependent motion between two or more particles 3. Calculate relative motion between two or more particles 4. Draw free-body diagrams and kinetic diagrams to represent the dynamic equilibrium of a particle using three types of coordinate systems 5. Write the equations of motion for a particle using three types of coordinate systems 6. Solve the equations of motion of a particle for forces or accelerations 7. Apply the principle of work and energy to a particle or a system of particles 8. Calculate power and mechanical efficiency of a machine 9. Apply the principle of linear/angular impulse and momentum for a particle or a system of particles, using conservation of linear/angular momentum when applicable 10. Calculate planar motion of a rigid body or system of rigid bodies undergoing pure translation, pure rotation and general planar motion 11. Calculate dependent motion in rigid body systems 12. Calculate the mass moment of inertia for a rigid body and system of rigid bodies 13. Draw free-body diagrams and kinetic diagrams to represent the dynamic equilibrium of a rigid body using three types of coordinate systems 14. Write the equations of motion for rigid bodies using two types of coordinate systems 15. Solve the equations of motion of a rigid body for forces or accelerations 16. Apply the principle of work and energy for a rigid body and system of rigid bodies, using conservation of energy when applicable 17. Apply the principle of linear/angular impulse and momentum for a rigid body and system of rigid bodies, using conservation of linear/angular momentum when applicable	
Education Method	This course follows a flipped class room approach. Material will be offered on Brightspace to prepare for each lecture. Additionally supervised self-study and open office hours with teaching assistants will be available to get help with the exercises.	
Literature and Study Materials	Engineering Mechanics: Dynamics by R.C. Hibbler + study pack plus Mastering Engineering with Pearson eText, 14th SI Edition, ISBN 9781292171951. Please note: Use this ISBN number only when ordering this book only. As otherwise your access code for Mastering Engineering will not be included. The use of Mastering Engineering is not required for this course but recommended. However, it is strongly advised students buy the Mechanics package from VSV Leonardo da Vinci as we have negotiated a special discount and special three-year access to Mastering Engineering. Buying loose copies will be more expensive and will only include a one-year access code. The pack contains the 3 Hibbeler books, with the studypack for Statics and Dynamics and an access code for Mastering Engineering for 36 months access.	
Assessment	Weekly quizzes that will be given indicative grades for feedback, but do not count towards the final grade (formative feedback). A list of recommended problems will be provided that can be used to practice for the weekly quizzes. Written final exam (with an entry requirement) covering all course learning objectives worth 10 points (100% of final grade). A resit exam opportunity (with an entry requirement) will be offered in the exam period of the subsequent quarter in which the course is offered. The resit will be worth 10 points (100% of final grade). Entry requirement for the final/resit exam: Students are eligible to take the final exam within the academic year in which a student has met the threshold for the number of required honest attempts for weekly quizzes. An attempt at a particular weekly quiz is seen as honest, if the paperwork that was submitted shows that a clear attempt was made to solve the problem(s) given in the particular weekly quiz. Students who did not meet the entry requirement for the final exam will be offered a repair opportunity for the resit exam. More details about the weekly quizzes and requirements for entry in the final exam and repair thereof will be explained in the first lecture of the course. All course material (book, lecture slides, assignments) will be in English. To avoid confusion between English and Dutch language terminology all relevant terms will be taught in English. Therefore all exams will be in English only. This includes the number format used (decimal point instead of comma, etc.).	
Permitted Materials during Tests	The final exam will be closed-book. Necessary formulas will be provided during the exam. The use of a pocket calculator is allowed. A calculator on a mobile phone, tablet or (laptop) computer is NOT allowed.	
Set-up	Students are encouraged to prepare the provided online content of the course prior to each of the lectures.	

The course material is divided into several chapters of the book (normally one per week). The weekly schedule consists of three lectures. In the lectures experiments will be performed and relevant example problems will be discussed. Moreover, time will be spent on discussing the results of the weekly assignments.

After each lecture, students are encouraged to undertake self-study either together or alone, solving recommended problems or attempting the practice test. During self-study directly after a lecture help from teaching assistants will be available. A representative list of problems will be provided for you to practice.

There will be open office hour sessions. During the open office hours you can ask help from the teaching assistants with the problems you've encountered. It is NOT compulsory to attend these open office hours. Open office hours will be held daily during the lunch break. Additionally, TAs will be available through the mail address Dynamics-AE@tudelft.nl. Details will be explained during the first lecture.

AE1205	Programming & Scientific Computing in Python for AE1	2
Responsible Instructor	Dr.ir. J. Ellerbroek	
Instructor	Dr. G.C.H.E. de Croon	
Contact Hours / Week x/x/x/x	0/0/0/8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Two 4 hour sessions per week (divided over 4 parts of the course , see below). Every session consists of instruction, examples, assignments and wrap-up/solutions. Also challenges and competitions are part of the course. Part 1 General Purpose Programming in Python Part 2 Scientific Computation & Plotting (NumPy, Matplotlib) Part 3 Numerical simulation, Visualisation & Animation (PyGame incl. game programming) Part 4 Contest & Diagnostic tests & Test Exam	
Study Goals	- translate a problem into an algorithm - basic programming skills - basics of Python+NumPy - numerical analysis - modeling - plotting data using matplotlib - simulation, visualisation & animation	
Education Method	Studio Classroom sessions, with: - Lecture (1 hour) - Exercises - (optional) assignments for 0.5 bonus point - (optional) competition for 0.5 bonus point - quizzes	
Computer Use	Bring your laptop with Python 3 installed	
Reader	Programming & Scientific Computing in Python for Aerospace Engineers	
Prerequisites	Install Python 3 (see Reader for instructions!)	
Assessment	Written/PC exam which contains multiple choice questions, assignments/open questions and fill in questions. Additionally one bonus point can be gained by meeting a certain quality in a final assignment, writing a simulation/game, and completing the session assignments. The bonus point is valid for the next four exams, so the exam and the resit of this year and the next.	
Exam Hours	0/0/0/3	
Permitted Materials during Tests	Non-programmable calculator or calculator in exam mode	
Set-up	Deliverables: week 1: Assignment week 2: Assignment week 3: Assignment week 4: Assignment week 5: Assignment week 6: project week 7: Assignment, test exam	

AE1222	Aerospace Design & Construction	9
Contact Hours / Week x/x/x/x	Project 0/0/14/14 Lecture 0/0/2/2 Studioclassroom 4 half days	
Education Period	3	
	4	
Start Education	3	
Exam Period	4	
	5	
Course Language	English	
Course Contents	This module deals with an introduction to Aerospace design.	
	The module is taught in two separate courses:	
	1) A project: Design and Construction	
	2) A course: Aerospace Design & Systems Engineering Elements	
Study Goals	At the end of the module students will be able to independently solve problems related to the design of aircraft and spacecraft.	
	For detailed learning objectives please see the learning objectives of the underlying courses.	
Education Method	The general method of teaching for this module is a combination of projects, active lectures and interactive practice sessions.	
	In the project, groups of approximately 8-10 students will act as a structural design team.	
	In the course, students are expected to take active part in class and during the studio classroom sessions.	
	For more details please see course descriptions	
Assessment	Each course will be assessed separately. See the course descriptions for more details. The final grade of the module AE1222 -	
	Aerospace Design and Construction(rounded to the nearest half) will be determined by the following formula:	
	grade AE1222 = {(grade AE1222-I)*5 + (grade AE1222-II)*4}/9	
	Each course grade (registered to one decimal place) must be 5.0 or higher AND the overall grade (rounded to the nearest half)	
	for AE1222 must be 6.0 or higher in order to pass this module.	
	No partial credits will be awarded for the individual courses of this module.	

AE1222-I	Design & Construction	5
Responsible Instructor	Dr.ir. O.K. Bergsma	
Instructor	C.D. Rans	
Instructor	E.J. van den Bos	
Instructor	Dr.ir. D.M.J. Peeters	
Instructor	I. Uriol Balbin	
Instructor	Ing. G.P.A. Wijnschenk Dom	
Instructor	D.M. Atherstone	
Contact Hours / Week x/x/x/x	Half days 0/0/14/14	
Education Period	3	
	4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In the project, groups of approximately 10 students will act as a structural design team. In the first project assignment, you will explore the properties of different materials and complete a preliminary design. In the second project assignment, you will design a structure that will be tested with a limited number of design variables; the structure has specified dimensions and should be able to sustain a specified load while minimizing the structural weight. In the final third project assignment, you will combine the lessons learned from the first and second assignment and apply them to a new (structural) design problem with more variables. Each assignment will have defined deliverables which need to be handed in before the stated deadlines during the project. A technical writing (TW) course runs concurrently with the project, but is scheduled separately. During the 5 TW sessions, you will be taught how to write technical reports and improve your writing skills. TW is run by TPM faculty staff. A Technical Writing Information Literacy (TWIL) course is part of the project. This (online) course will teach you searching for relevant literature which is required to be incorporated into the Design Reports for each assignment. This course is part of technical writing. This course is run by TU Delft library staff. Information with respect to course logistics, relevant deadlines and project assignments will be communicated using the Brightspace course page. Note: First year students are automatically enrolled and assigned to a (new) group. Upper year students who did not complete the Design and Construction project please contact Education-AE (Education-AE@tudelft.nl) to be registered before February 1st.	
Study Goals	At the end of the project you will be able to: - Independently complete a full design cycle of an AE (sub)system - Performing a complex team project under external guidance and reflect on its outcome - Write an effective design report	
Education Method	Students will work in a group with approximately 10 members for two 4-hour project sessions per week throughout the entire semester. The project assignments are defined and supervised by academic staff from the Aerospace Faculty assisted by teaching assistants. Two groups have a responsible teaching assistant (TA). Progress will be monitored by project meeting (at the start and end of the session) with the TA. Technical Writing (TW) is a separate course, but taught and assessed by the technical design reports which are required to be produced as part of the Design and Construction project, taught 5 x 2 hours during the semester. Two groups will have TW teacher and coached. Before TW sessions a literature study online test needs to be completed (TWIL). Details on TWIL and a TW schedule will be posted on Brightspace at the start of the project. If circumstances dictate project session(s) format can be changed by the responsible instructor at any time. More details and information will be available on Brightspace.	
Computer Use	As reports need to be produced, you will need a laptop with internet capability. If circumstances dictate students are required to have an electronic device (laptop, desktop or tablet) with a camera and (built-in) microphone in order to participate in the project. The camera and microphone should be active during project session to allow for communication and group work. Students are also responsible for ensuring that they have access to reliable Wi-Fi/internet connection.	
Literature and Study Materials	Mandatory: Brightspace course page AE1222-I - Design and Construction, project reader(s), video's, supplement materials posted on Brightspace. NOTE: a new reader will be made available at the beginning of each project assignment. As the assignment changes each year, previous reader(s) is no longer valid!	
Prerequisites	Knowledge from BSc 1 courses given in Q1, Q2 and Q3 (at the end of the project)	
Assessment	The final grade for the project consists of the following partial grades: 1) A group grade for the second project assignment (Q3) - (15%) 2) A group grade for the third project assignment (Q4) - (35%) 3) A grade for technical writing (individual deliverable (5%) and group design report (15%)) - (Total 20%) 4) An individual grade - (30%) Note (1) and (2): The group grades for project assignment 2 and 3 are assessed on technical report content (not writing) by Staff and TA's. Note (3): The technical writing individual grade will be assessed with an independent individual deliverable. The final report of assignment 3 is assessed on writing by the technical writing staff for the group grade. Technical Writing Information Literacy (TWIL) course needs to be completed before the set deadline to obtain a passing grade in this section. Note (4): This grade is based on project peer reviews, attendance and work ethics for the complete project duration (Q3 and Q4). During the project several deadlines, for peer evaluation and assignments/deliverables, need to be met. Failing to meet a project deadline will result in a grade penalty or failing the entire course. Grades will be determined per the requirements listed below: 1. The whole project has been completed 2. All rules regarding absence are met	

3. All compensatory assignments have been completed successfully (if applicable)
4. No more than one partial grade is lower than 6.0
5. All partial grades are 5.0 or higher

If any of the conditions listed under 1-5 are not met, a final grade of 1.0 is awarded for the project. If a grade of 1.0 is awarded to a student, the entire project must be redone in the following academic year.

Both partial and final grades will be rounded to 1 decimal.

The following rules regarding absence hold:

1. The student must attend the first week of the project. Not being present in the first week of the project results in exclusion from participation in the project in that academic year.
2. Students must attend all scheduled project session, but due to illness or family circumstances are allowed to miss a maximum of 2 project sessions per period (half semester).
3. Students are also allowed, given sufficient communication with teaching assistants, staff and team members, to join one project session online in total. Joining online is only allowed via the Microsoft Teams application, using the dedicated channel. Their on-camera online presence must be traceable by the teaching assistant, teammates, and staff. No online attendance is permitted for practical sessions.
4. If a project has a supporting course the student is only allowed to be absent for one session of this course. This session is not counted as a missed project session as mentioned under 4.
5. Attendance (online/on-campus) is mandatory during the scheduled project session hours. Missing time (either by being late or leaving at any time) during project sessions can result in being registered as missed sessions.
6. Missed sessions must be compensated according to the rules stated below.
7. Absence for whatever reason for more than the allowed number of sessions or failure to deliver a compensating assignment will mean that the student cannot obtain a passing grade for the project.
8. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counselors for guidance.

For compensating absence, read the information under Special Information

Enrolment / Application

First year students are automatically enrolled and assigned to a(new) group. Upper year students who did not complete the Design and Construction project please contact Education-AE (Education-AE@tudelft.nl) to be registered before February 1st.

Special Information

Continuation Assessment

For compensating absence, the following applies:

1. The student is responsible for fulfilling the compensatory assignment, at a time or within a time limit set by the responsible lecturer (this also applies to supporting courses).
2. The quality of the compensatory assignment is assessed by the responsible lecturer. If the student does not complete the assignment within the allocated time and/or with sufficient quality, they will fail the project.
3. Compensatory assignments cannot be amended or redone if deemed unsatisfactory upon assessment by the responsible lecturer and results in failing the project.
4. No opportunities are offered outside the duration of the project for making up for missed sessions.
5. Group work (working together) on individual assignments is not allowed, each student must hand in original work in your own words.
6. The assignments must be made in the English language.
7. The assignments may NOT be handwritten (hand sketches are allowed).
8. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counselors for guidance.

Set-up

Deliverables:

Period 3:

- Assignment 1 (Design Report)
- Assignment 2 (Design Report and Design Drawing)

Period 4:

- Assignment 3 (Design Drawing)
- Assignment 3 (Design Report)

AE1222-II	Aerospace Design & Systems Engineering Elements	4
Responsible Instructor	Dr.ir. R. Vos	
Responsible Instructor	Dr. J.A.M. Vanhamel	
Responsible Instructor	I. Akay	
Instructor	Dr.ir. M.F.M. Hoogreef	
Contact Hours / Week x/x/x/x	0/0/2/2 + 2 x 2 hr studio classroom sessions in Q3 and 4 x 2 hr studio class room sessions in Q4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Course Contents	This course addresses the design of aerospace systems in general and, more in particular, the preliminary design of airplanes, spacecraft, and rocket launch vehicles. The following topics are addressed: Aerospace design: - Design process (in general and with application to airplanes, spacecraft, and launch vehicles) - Role of SE in the design process - Functional & operational analysis, requirements generation - Design concept selection & description Airplane design: - Design requirements - Analysis of aircraft configurations - Fuselage design - Preliminary weight estimation - Preliminary sizing of wing area and engine power/thrust - Wing and propulsion system design and integration - Tail and landing gear design and integration Spacecraft design: - Spacecraft mission, main functions, main requirements, vehicle concepts - Preliminary sizing of the spacecraft subsystems: 1) structures & mechanisms, 2) thermal control, 3) electrical power generation, 4) propulsion, 5) attitude determination and control, 6) command and data handling and 7) telecommunications Launch vehicle design: - Launch vehicle mission, main functions, main requirements - Launch vehicle concepts - Single- and multi-staged launch vehicles - Preliminary sizing of rocket stages and launch vehicle subsystems The material treated in this course will be applied in later thematic projects (BSc year 2) and the Design Synthesis Exercise (BSc year 3). Follow-up courses are scheduled in years 2 and 3 of the aerospace engineering bachelor curriculum.	
Study Goals	At the end of this course, the student shall be able to: - Identify and explain the aircraft/spacecraft architecture and the conceptual design process. - Identify and explain aircraft and spacecraft missions - Perform a simple functional analysis and generate a list of requirements - Generate a conceptual aircraft design and perform a preliminary sizing of the aircraft including wing, fuselage, tail, and landing gear. - perform a preliminary design & sizing of the most important spacecraft and space launch vehicle subsystems including the subsystems structures and mechanisms, thermal control, electrical power generation, propulsion, attitude determination and control, command and data handling and telecommunications	
Education Method	Lectures and tutorials The course consists of a series of 14 lectures supported by a number of in-class training sessions (tutorials), divided over two quarters. In the lectures the main topics are introduced and explained supported by visual aids. The tutorials provide the student the opportunity to actively participate in the learning process by conducting design-oriented exercises. Teaching staff and teaching assistants will be available for providing guidance and for discussion. Next to the lectures and tutorials, feedback exercises are made available via Brightspace, such that the student can practice. Subjects per lecture: Part 1 (3rd quarter) 1 - Design process and requirements 2 - Analysis of aircraft configurations 3 - Fuselage design 4 - Preliminary weight estimation 5 - Preliminary sizing of wing area and engine power/thrust 6 - Wing and propulsion system design and integration 7 - Tail and landing gear design and integration Part 2 (4th quarter): 1 - Spacecraft design process and requirements, main functions, design options & initial sizing 2 - Spacecraft initial sizing, structures and mechanisms 3 - Thermal control, electrical power generation 4 - Propulsion, attitude determination and control 5 - Command and data handling, telecommunications 6 - Introduction to launcher design and concept analysis 7 - Launcher design details with focus on the launcher subsystems Tutorials part 2 (4th quarter): Tutorial 1: Initial sizing of spacecraft based on historical data using simple relations. Tutorial 2: Initial sizing of the subsystems 1) Structures and mechanisms, 2) Thermal control and 3) Electric power generation Tutorial 3: Initial sizing of the subsystems 4) Propulsion, 5) Attitude determination and control, 6) Command and data handling and 7) Telecommunications Tutorial 4: Initial sizing of launch vehicles and their subsystems	
Literature and Study	- R. Vos, Introduction to Airplane Design, unpublished textbook manuscript.	

Materials	- B. Zandbergen Aerospace Design & Systems Engineering Elements I, parts: Spacecraft Design and Sizing and Launch Vehicle Design and Sizing.
Assessment	Assessment for the course consists of the assessment of a) the airplane design part and b) the spacecraft design part: Ad a) The assessment of the airplane design part is based on a take-home exam. This exam consists of 7 take-home assignments that are individually tailored to each student. The take-home exam is open book and needs to be submitted in a standardized format (template) that is provided by the instructor(s). The submission deadline for the exam is near the end of Q3. There is a resit for the take-home exam, for which the deadline is near the end of Q4. Ad b) The assessment of the spacecraft design part is based on an exam at the end of the course and lasts 90 minutes. In addition, there is a resit exam for the spacecraft design part, provided in the fifth exam period. Grading Rules: The grading is based on the students results of the two parts (a and b). At the end of the semester, the student has 2 partial grades, each on a scale from 1 to 10: - Airplane design take-home exam, a - Spacecraft design exam, b Each partial grade a and b must be 4.0 or higher to pass this course. The final grade is computed as follows (parts a and b contribute equally to the final grade): Final grade $= (a+b)/2$ Note: Partial grades do not transfer to the subsequent academic years.
Permitted Materials during Tests	Only the standard BSc calculator (Texas Instruments (TI) 30xb multiview) is allowed during the written exam.

AE1241	Physics	6
Responsible Instructor	Dr.ir. M.I. Gerritsma	
Instructor	Dr. F. Yin	
Instructor	Dr. R. Merino Martinez	
Instructor	A. Bombelli	
Instructor	Dr. J.J.D. Loicq	
Contact Hours / Week x/x/x/x	0/0/4/4	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Required for	This is a basic course in classical physics and, like courses in mathematics, it provides indispensable and relevant basic knowledge for the aerospace engineer.	
Parts		
Course Contents	The course is based on the chapters 14-35 of the book Physics for Scientists & Engineers (with Modern Physics), Fifth Edition, Pearson, by Douglas C. Giancoli. For the course Pearson made a customized edition called "Thermodynamics, waves and electromagnetism" containing these chapters only. The book contains a code which is mandatory for participation in the practical sessions. Therefore a new book is required. Waves part (chapters 14-16): - simple harmonic motion, energy of harmonic motion, simple and physical pendulum, damped harmonic motion, forced oscillation (resonance) - wave motion (transverse and longitudinal), energy transported by waves, wave equation, superposition, reflection, transmission, interference, refraction (Snell's law), diffraction, standing waves (resonance) - sound, sound level (decibel scale), vibrating strings and air columns, quality of sound, noise, superposition, interference (beats), Doppler effect Thermodynamics part (chapters 17-20): - temperature, thermometers, thermal equilibrium, zeroth law of thermodynamics, thermal expansion, gas laws (Boyle, Charles and Guy-Lussac), absolute temperature, ideal gas law, ideal gas temperature scale (triple point), Avogadro's number - kinetic theory of gases, molecular interpretation of temperature, Maxwell distribution of molecular speeds, real gases and changes of phase, critical temperature, phase diagram, triple point, vapor pressure and humidity, Van der Waals equation of state, mean free path, diffusion (Fick's law) - heat as a transfer of energy, internal energy (of ideal gas), specific heat, calorimetry, open, closed and isolated systems, latent heats, first law of thermodynamics, work, state variables, work done for isothermal, adiabatic, isobaric and isovolumetric processes, molar specific heat of a gas (at constant pressure and constant volume), principle of equipartition of energy, equation of state for adiabatic expansion/compression of an ideal gas, heat transfer by conduction, convection and radiation (Stefan-Boltzmann law) - Clausius statement of the second law of thermodynamics, heat engines, Kelvin-Planck statement of the second law, reversible and irreversible processes, Carnot cycle, Carnot efficiency, Otto cycle, heat pumps, refrigerators and air conditioners (coefficient of performance), entropy and why it is a state variable, general statement of the second law of thermodynamics (in terms of entropy), order and disorder, statistical interpretation of entropy (Boltzmann formula), microstates and the macrostate of a system, third law of thermodynamics, thermal pollution, energy resources and global warming Electromagnetism part (chapters 21-31): - electrostatics, Coulomb's law, electric field, electric dipole, electric flux, Gauss's law, electric potential, equipotential surface, capacitance, capacitors in series and in parallel, electric energy storage, dielectrics - electric battery, electric current, resistance (Ohm's law), resistors in series and in parallel, electric power, DC circuits, Kirchhoff's rules, RC circuits - magnetism, force on electric current and electric charge in magnetic field, torque on a current loop, magnetic field due to straight wire, Ampère's law, Biot-Savart law, magnetic materials, electromagnets - electromagnetic induction, Faraday's law, Lenz's law, electric generators, transformers - mutual and self inductance, electromagnetic oscillations, AC circuits containing resistance, inductance and capacitance (resonance), impedance matching - above mentioned laws of electromagnetism are revisited in the form of Maxwell's equations, Gauss's law for magnetism - electromagnetic (EM) waves as solutions of Maxwell's equations, light as EM waves, EM spectrum, speed of light, Poynting vector, radiation pressure, wireless communication	
Course Contents Continuation	Optics part (chapters 32-35): - The ray model of light, reflection; image formation by a plane mirror, formation of images by spherical mirrors, concave and convex mirrors, index of refraction, Snell's law, visible spectrum and dispersion, total internal reflection, fiber optics, refraction at spherical surfaces. - Thin lenses, ray tracing and focal length, the thin lens equation, lensmaker's equation, cameras, film and digital, the human eye, magnifying glass, telescopes, compound microscope, aberrations of lenses and mirrors. - Waves vs particles, Huygens principle and diffraction, the Law of refraction, mirages, interference, Young's double slit experiment, interference in thin films, Michelson interferometer, polarization, liquid crystal displays (LCD), scattering of light by the atmosphere, brightness, lumens and luminous intensity, efficiency of light bulbs. - Diffraction by a single slit or disk, intensity of single slit diffraction pattern, diffraction in the double slit experiment, interference vs diffraction, limits of resolution, circular apertures, resolution of telescopes, the Rayleigh limit, resolution of the human eye and useful magnification, diffraction grating, spectrometer and spectroscopy, peak widths and resolving power for diffraction grating, X-rays and X-ray diffraction, X-ray imaging and Computed Tomography (CT Scan), microscopes and contrast.	
Study Goals	At the end of this course the student will be able to use physics to solve basic problems from science and engineering (and in particular aerospace engineering) related to thermodynamics, waves, electromagnetism and optics.	
Education Method	- Lectures explaining the theory and some worked-out problems from the book by Giancoli - Self-study of the theory and practicing with the worked-out examples and problems from the book - Practicing with the problems in the tool "Mastering Physics" (for each chapter a mastering physics homework test is available)	
Literature and Study Materials	- Chapters 14-35 of the book Physics for Scientists & Engineers (with Modern Physics), Fifth Edition, Pearson, by Douglas C. Giancoli. Pearson made a customized edition called "Thermodynamics, waves and electromagnetism" containing these chapters only - Homework computer tool "Mastering Physics"	

Prerequisites	First-year calculus
Assessment	Two written exams (multiple-choice and open questions). The first exam is about thermodynamics, waves and electrostatics/electrodynamics, the second exam is about electromagnetism and optics. No formula sheet will be given for the exams. The grade for this course is the average of the two marks obtained for these two partial exams provided that the grades for both partial exams is at least a 5. It is possible to retake the partial exams independently. Transition AE1240 -> AE1241 1. The course content will no longer be offered in two separate courses within a module but offered as one single course. 2. For students from the 2021 2022 cohort and earlier who have not yet completed AE1240-I and/or AE1240-II, the following applies: a. In 2022 2023, two opportunities will be offered to complete AE1240-I and/or AE1240-II. b. Students can also follow the new course which will be updated in OSIRIS/MyTUDelft automatically. c. If AE1240 is not completed in 2022 2023, students will need to complete AE1241 instead. d. Partial results for AE1240-I and AE1240-II will no longer be valid after 31st August 2023.
Permitted Materials during Tests	Only the calculator provided by the Faculty can be used.

WI1402LR	Calculus II	5
Responsible Instructor	Dr. M.B. van Hoven	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Summary	Multivariable Calculus	
Course Contents	The goal of the course is to lay a strong foundation for the numerous and various applications of calculus, both directly within the field of aerospace engineering as in advanced engineering mathematics. The course focusses on both differentiation and integration techniques involving multivariable functions and vector fields. The main subjects of the course are: Techniques for partial and directional derivatives (tangent plane, linearisation, gradient vector, extreme values,) Integration techniques for functions of two or more variables. Double integrals using polar coordinates. Triple integrals using cylindrical and/or spherical coordinates. Line and surface integrals of functions and vector fields. Theorems of vector calculus by Green, Stokes and Gauss.	
Study Goals	At the end of this course, the student masters the course subjects at a variety of levels: Knowledge: reproduce and name the learned definitions, theorems and formulas. Comprehension: explain, interpret and derive the learned theorems and formulas. Application: apply the learned definitions, theorems and formulas in concrete mathematical problems. Analysis: decompose mathematical problems in relevant smaller (easier) subproblems. Synthesis: Composing strategies for solving a mathematical problem by combining several learned techniques. Evaluation: use learned mathematical tools to critically asses (and judge on) a given idea or proposition	
Education Method	A threefold workflow around each lecture: Preparation: Students study a pre-lecture video and/or do pre-lecture exercises to prepare for class. Participation: Students come to the lecture. An active attitude is required: taking part in practice exercises and poll questions posed by the lecturer. Asking questions. Practice and study: After the lecture the students practice exercises from the book and in Grasple. Concepts discussed in class should be studied in detail (use the book).	
Literature and Study Materials	Adams, R.A. , "Calculus, a complete course", 10th ed., Addison Wesley (2021), ISBN 9780135732588	
Assessment	Written exam (two hours) at the end of the third quarter. The assessment contains short answer questions (only answer assessed) and open questions (calculation/motivation assessed) being a representative selection of the entire course content.	
Remarks	It is not allowed to use electronic devices such as calculators, mobile phones and laptops/notebooks etc. during the exam. Written notes and formulae sheets are also prohibited.	
Co-Instructor	Dr. W.M. Schouten-Straatman	

WI1403LR	Linear Algebra	5
Responsible Instructor	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	In this course we will study the properties of matrices and vectors. These results will then be used to solve systems of linear equations, describe linear transformations, determine projections, obtain least squares solutions of systems of equations and solve systems of linear differential equations.	
Study Goals	At the end of this course the student is able to... 1. solve systems of linear equations, 2. identify whether a system of linear equations is solvable and whether the solution is unique, 3. apply the equivalence between a vector equation, a matrix equation and a system of linear equations, 4. give a geometric description of solutions sets of vector equations, 5. determine the linear (in)dependence of a set of vectors and identify the (geometric) properties of linear dependence, 6. check whether a transformation is linear and determine the standard matrix of a linear transformation, 7. perform matrix operations (sum, scalar multiple, multiplication, transpose), describe their properties and apply these properties, 8. describe the properties of invertible matrices, determine the inverse of a non-singular matrix and apply these topics, 9 use the definitions of subspace, column space and null space of a matrix and determine a basis for a subspace, 10. apply the concept of dimension of a linear subspace, rank of a matrix and the rank theorem, 11. determine the determinant of a matrix and apply the properties of a determinant, 12. calculate and apply properties of the inner product in the context of orthogonality and orthogonal sets, 13. compute the orthogonal projection on a subspace, 14. determine an orthogonal basis of a subspace by means of the Gram-Schmidt process, 15. use the least-squares method and apply the least-squares method to linear models, 16. determine the eigenvalues and eigenvectors of a matrix, 17. determine whether a matrix is diagonalizable and if so, determine a diagonalization of the matrix, 18. analyse matrices with complex eigenvalues and eigenvectors, 19. solve a system of linear differential equations and sketch its trajectories, 20. determine whether a statement about the aforementioned topics is true or false and either prove or disprove that statement.	
Education Method	Lectures blended with exercise sessions and online homework exercises.	
Literature and Study Materials	Open Linear Algebra Book. Hensbergen, Verhulst et al.	
Assessment	The exam is a written exam.	
Permitted Materials during Tests	The use of the standard bachelor calculator (Texas Instruments (TI) 30xb multiview) is allowed.	

WI1421LR	Calculus I	6
Responsible Instructor	Dr. M.B. van Hoven	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Summary	Single Variable Calculus	
Course Contents	The goal of the course is to lay a strong foundation for the numerous and various applications of calculus, both directly within the field of aerospace engineering as in advanced engineering mathematics. The possibility of using computer algebra systems allows for a shift in emphasis from technical skills towards understanding the underlying concepts.	
Study Goals	Course subjects: - Vector arithmetic (addition, scalar multiplication, dot product, cross product). - Differentiation, linearisation, differentials. - Integration techniques for functions of one variable (e.g. substitution rule, integration by parts). - First-order differential equations by analytical methods. - Complex numbers, both in Cartesian and polar form (addition, multiplication, division, powers, etc.). - Second-order linear differential equations with constant coefficients. - Series and sequences: Convergence tests and techniques. - Taylor and Maclaurin series for the approximation of functions. - Introduction to multivariable functions: continuity and differentiability. At the end of this course, the student masters these subjects at a variety of levels: Knowledge: reproduce and name the learned definitions, theorems and formulas. Comprehension: explain, interpret and derive the learned theorems and formulas. Application: apply the learned definitions, theorems and formulas in concrete mathematical problems. Analysis: decompose mathematical problems in relevant smaller (easier) subproblems. Synthesis: Composing strategies for solving a mathematical problem by combining several learned techniques. Evaluation: use learned mathematical tools to critically assess (and judge on) a given idea or proposition	
Education Method	Lectures + seminars Typical routine: - Preparation: Students do homework, study a pre-lecture video (or do pre-lecture exercises) - Participation: Students come to the lecture which is a block of two hours. In the lecture a new topic will be introduced by the teacher. During the lecture there will be time to do exercises. Often a practical application of the theory is discussed. - Post-lecture work: Students do homework assignments. These contain book exercises as well as exercises in a digital environment.	
Literature and Study Materials	Adams, R.A. , "Calculus, a complete course", 10th ed., Addison Wesley (2021), ISBN 9780135732588	
Assessment	Preliminary examinations (two hours) at the end of the first and second quarters. Both assessments have a two-hour duration and consist of a number of short answer questions (only answer is assessed) and a number of open questions (calculation/motivation is assessed). *In the event of on-campus restrictions (e.g. due to COVID), assessments will be 90 min online (Grasple) proctored exams.	
Special Information		
Remarks	At the exam, it is not allowed to use electronic devices (such as calculators, mobile phones and laptops/notebooks etc.), formulae sheets.	
Co-Instructor	Dr. N.D. Verhulst	

Year	2024/2025
Organization	Aerospace Engineering
Education	Bachelor Aerospace Engineering

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AE2111		Aerospace System Design	8
Responsible Instructor	Prof.dr. S.J. Watson		
Contact Hours / Week x/x/x/x	see AE2111-I and AE2111-II		
Education Period	1 2		
Start Education	1		
Exam Period	1 2		
Course Language	English		
Course Contents	The module AE2111 consists of the project AE2111-I (System Design) and the course AE2111-II (Aerospace Design and Systems Engineering Elements II) For detailed content please consult the individual parts of the module		
Study Goals	The module aims at teaching students to identify, analyze and design aerospace (sub)systems and to report this newly acquired knowledge either as an individual by taking an exam, doing a test or presenting it with visual aids or as part of a group by working on a design and handing in work package reports.		
Education Method	For the education methods, please consult this description of the individual parts of the module		
Assessment	Taking part in the module System Design will result in two separate grades, one for the course AE2111-II and one for the project AE2111-I, each with their own assessment method. The grades will be rounded off to 0.1 points. The final grade consists of 37.5% of the course grade and 62.5% of the project grade and will be rounded off to the nearest 0.5 point. To pass, the final grade must be a 6 or higher.		

AE2111-I	Systems Design	5
Responsible Instructor	Prof.dr. S.J. Watson	
Instructor	Dr.ing. S. Giovanni Pereira Castro	
Instructor	Dr.ing. A. Heidebrecht	
Instructor	Dr. J.A.M. Vanhamel	
Instructor	I. Akay	
Contact Hours / Week x/x/x/x	8/8/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The theme of the project is System Design. As a subject of this project, teams work on either the design of a spacecraft, or the design of an aircraft wing. This project is supported by a course on 'Oral Presentations'.	
Study Goals	The course aims at providing a learning experience for students, in which they apply the (theoretical) content of thematic courses in a practical, project setting. The students are organized in groups working on real-life problem cases, resulting in a professional output product: a number of work package reports and technical presentations. At the end of this project-based course, student will be able to - Apply knowledge and practice the methods from AE courses for the design of (sub)systems; - Determine and describe design options; - Evaluate the performance of systems, describe trade-offs and reflect on the selected design choices; - Use (online) communication tools to effectively co-operate in a team; - Prepare and give an oral presentation on a technical subject, using effective visual aids.	
Education Method	Students will work on a project in groups of up to 10 members, for two 4-hour sessions per week, throughout both quarters Q1 and Q2. The projects are defined and supervised by academic staff from the Aerospace Faculty, assisted by Teaching Assistants (every two groups have a responsible TA). Progress will be monitored by frequent project meetings with the TA. Different 'aircraft-wing' projects and 'spacecraft missions' will be distributed amongst the groups. The accompanying 'Oral Presentations' course contains a number of group practice sessions, which are aimed at creating presentations on the project results during the mid-term and final reviews.	
Books	None - all required material is provided during the course.	
Prerequisites	The project entrance requirements are: 45 ECTS of the first year of the BSc. Aerospace Engineering programme (BSA), including the first-year project AE1111-I.	
Assessment	The students are assessed on 3 different project components: - An individual grade for job performance, academic and project skills (20%); - A group grade for the project work package reports (60%); - An individual mark for the 'Oral Presentations' course, given by the course lecturers (20%).	
Special Information	Attendance: Attending the scheduled project sessions is obligatory and will be registered by the responsible teaching assistant. 1. Students are allowed to miss a maximum of 2 project sessions per period (half a semester), for the project itself. 2. If a project has a supporting course the student is only allowed to be absent for one session of this course. This session is not counted as a missed project session as defined under 1. 3. Missing time (either by being late or leaving at any time) during project sessions can result in it being registered as a missed session. 4. Missed sessions must be compensated according to the rules stated below. 5. The student must attend the first week of a project. Not being present in the first week of the project results in exclusion from participation in the project for that academic year. 6. A student is not allowed to miss three successive sessions in the timeframe formed by the last two weeks of the 1st period and the first two weeks of the 2nd period. 7. Absence for whatever reason for more than the allowed number of sessions or failure to make up for missed hours will mean that the student cannot obtain a pass grade for the project. 8. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counsellors for guidance. For compensating absence, the following applies: 1. The student is responsible for fulfilling the compensatory assignment, at a time or within a time limit set by the responsible lecturer (this also applies to supporting courses). 2. The quality of the compensatory assignment is assessed by the responsible lecturer. If the student does not complete the assignment within the allocated time and/or with sufficient quality, they will fail the project. 3. Compensatory assignments cannot be amended or redone if deemed unsatisfactory upon assessment by the responsible lecturer and results in failing the project. 4. No opportunities are offered outside the duration of the project for making up for missed sessions. 5. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counsellors for guidance. Students will only be awarded a final grade for the project if: 1. The whole project has been completed and all compensatory assignments have been completed successfully and, 2. No more than one grade is lower than 6.0 and, 3. All grades are 5.0 or higher. The project grade will be rounded to 1 decimal point. The final project grade must be 6.0 or higher to pass the project. If a student fails the project they have to re-register to redo the project during the following academic year. If a student is required to resit the Oral Presentation element of the course, their resit grade for this element will be capped at 7.0.	
Set-up	Deliverables: Quarter 1: week 2: Work package report 1 week 5: Work package report 2 Quarter 2:	

week 1: Work package report 3
week 2: Mid term presentation
week 5: Work package report 4
week 7: Work package report 5
week 8: Final presentation

AE2111-II	<i>Aerospace Design & Systems Engineering Elements</i>	3
Responsible Instructor	A. Cervone	
Instructor	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	General Subsystems a. What are the vehicle subsystems and how do they function and interact? Key Subsystems b. How to design an operational reliable spacecraft communications system and an ADCS system? c. How to determine the wing geometry according to the aerodynamic requirements? How to design the high lift devices and the ailerons of an aircraft? Which Class II weight estimation methods are available, and how can they be used?	
Study Goals	1. Identify vehicle subsystems and understand their functionality 2. Analyze most important subsystems 3. Design top-level subsystems including their operations 4. Identify some of the main subsystems of Aircraft and Spacecraft and explain their functions 5. Describe Space flight control lay-outs and Attitude Determination and Control System (ADCS) designs and show their functionality 6. Understand how to estimate the lift and drag characteristics of both an airfoil and a wing. 7. Understand the conceptual sizing approach of aircraft high lift devices and ailerons 8. Understand how to estimate the weight of each aircraft component through Class II weight estimation methods 9. Explain and apply basic rules to design and size a spacecraft communications system 10. Conceptually design an aerospace sub-system (e.g. a high lift devices system) or mission element/spacecraft bus (e.g. the ADCS sub-system) 11. Explain and analyze redundancy concepts and systems as part of risk and reliability engineering	
Education Method	Lectures, individual written exam and group tutorials	
Literature and Study Materials	- Lecture Slides - Lecture notes Aircraft Design, Gianfranco La Rocca - Space Mission Analysis and Design, James R. Wertz and Wiley J. Larson (optional study book, NOT mandatory to pass the course)	
Assessment	- One exam, written and/or digital (counting for 2/3 of the final course grade), including 50% questions on the spacecraft part and 50% questions on the aircraft part - Two group tutorials (each counting for 1/6 of the final course grade), one on the aircraft part and one on the space part - A retake will be offered for the exam and the two tutorials - All three assessment items (exam + aircraft tutorial + spacecraft tutorial) are mandatory to pass the course: no participation or insufficient grade in one of them will result in a course fail - More details on assessment and pass criteria are given in the introduction lecture	
Remarks	This course is the second in a series of three. It is preceded by Aerospace Design and Systems Engineering Elements I in BSc year 1 and followed by Systems Engineering and Aerospace Design in BSc year 3.	
Set-up	The course will apply similar learning methods for spacecraft and aircraft. It will use Systems Engineering methods as an integrative element for the two application areas of aircraft and spacecraft. The teaching elements are two blocks using active lecturing, each with 10 lecture hours of 45 minutes, one block on spacecraft, one on aircraft in addition to one introduction lecture hour. Deliverables: two homework tutorials (one on space, one on aircraft) + final exam.	

AE2130	Aerodynamics Sub- and Supersonic	7
Module Manager	Dr.ir. M.I. Gerritsma	
Contact Hours / Week x/x/x/x	4/6/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2 3	
Course Language	English	
Course Contents	Subsonic and Supersonic Aerodynamics + Wind tunnel practical	
Study Goals	Obtaining a good knowledge in the fundamentals of aerodynamics and its relevance to and application in the aerospace technology domain	
Education Method	lectures + wind tunnel practical	
Literature and Study Materials	1. Text Book: Fundamentals of Aerodynamics, John D. Anderson 2. Extra material 3. Wind Tunnel Practical Manual	
Assessment	written exams + report for WT practical	
Tags	Calculus Challenging Fluid Mechanics	

AE2130-I	Aerodynamics I	3
Responsible Instructor	A. Sciacchitano	
Instructor	Dr.ir. T. Michelis	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Derivation of basic flow equations Fundamental concepts of fluid flow Potential flows in two and three dimensions; elementary solutions Airfoil and wing theory for inviscid, incompressible flow Determination of lift and induced drag of finite wings	
Study Goals	At the end of the course the student should be able to: 1.Explain aerodynamic concepts and apply aerodynamic theory for incompressible flows. 2.Apply the fundamental concepts of mechanics to describe fluid flow 3.Derive the governing equations that are valid under general conditions (viscous, compressible flow) 4.Describe the relevance of different flow regimes (compressible/incompressible, viscous/inviscid) 5.Describe and apply the aerodynamic concepts that are used in the description of flows (streamlines, circulation, vorticity, etc.) 6.Explain the essential phenomena that occur in incompressible flows in relation to the generation of forces (lift, drag) 7.Calculate the lift and pressure distribution over airfoils using potential theory (thin-airfoil theory) 8.Indicate the effects of viscosity on the behavior of airfoils 9.Apply lifting-line theory and VLM for calculations of wing lift, lift distribution and induced drag 10.Apply the airfoil & wing theory in a wind tunnel experiment	
Education Method	Lectures and homework assignments	
Literature and Study Materials	- Text Book: Fundamentals of Aerodynamics, John D. Anderson (5th ed. 2011, or successive), Chapters 1 - 6 - Course notes for some additional topics (available on Brightspace)	
Assessment	Written, open questions.	
Permitted Materials during Tests	Normal calculator (non-programmable)	
Set-up	Throughout the semester, students are expected to study the currently discussed material, and to do preparatory homework assignments. Within the lectures, the instructor discusses relevant exercises, and presents new material. In the evening, the student reviews the lecture material covered during the day, and begins on the next homework assignment.	
	Following the completion of the theoretical component of the course, students perform a wind tunnel practical (AE2130-II).	

AE2130-II	Low Speed Wind Tunnel Test	1
Responsible Instructor	M. Kotsonis	
Instructor	M. Li	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Expected prior knowledge	P examination is recommended.	
Parts	1) Pre-test lecture 2) One half day performing the exercise. 3) Three half days to work out the results and to prepare a report.	
Course Contents	1) Measuring the properties of a two-dimensional airfoil and a three-dimensional wing in a low speed windtunnel 2) Pressure measurements, wake measurements and demonstration of flow phenomena. 3) The measurement results are compared to basic CFD calculations on the spot.	
Study Goals	1) Illustration of some of the topics addressed in the course ae2110 2) Getting acquainted with the real flow conditions (e.g. effects of viscosity) over a 2 airfoil and 3D wing 3) Illustration of the restricted accuracy of basic CFD approaches for wing aerodynamics	
Education Method	1) Pre-Lab lecture 2) Windtunnel experiment + CFD analysis. Comparison of methods and reporting.	
Literature and Study Materials	Hand-outs	
Assessment	Assessment of lab report, grading with a mark.	
Set-up	The practical exercise AE2130-II is given once every year in the second period. The exercise takes place in the low speed windtunnel, LTT, Leeghwaterstraat 42, Delft Deliverables: final report three weeks after the completion of the lab exercise	

AE2130-III	Aerodynamics II	3
Responsible Instructor	Dr.ir. F.F.J. Schrijer	
Instructor	Dr.ir. W.J. Baars	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	AE2130-I Aerodynamics 1 AE1241 Physics WI2180LR-I Differential equations AE1205 Programming & Scientific Computing in Python for AE1 (needed to process the data from the (optional) wind tunnel practical)	
Course Contents	How to apply the basic laws of mechanics and thermodynamics to describe compressible flows? What are the essential phenomena that are relevant and specific to aerodynamic behaviour under compressible flow conditions How to apply the theory in the prediction and computation of compressible flows (airfoils, engine intakes, nozzles)	
Study Goals	At the end of this course, the student will be able to: 1. Understand aerodynamic concepts and apply aerodynamic theory for compressible flows. 2. Explain which essential phenomena occur in compressible flows and explain the relevance of different flow regimes (transonic, supersonic, hypersonic) 3. Apply the fundamental equations of fluid mechanics and thermodynamics to describe compressible flows; derive the governing equations for compressible flow 4. Calculate the flow through channels and nozzles. 5. Calculate the flow properties of airfoils in supersonic flows (linearized theory, shockwave-expansion theory). 6. Explain the effects of viscosity on the behavior of airfoils in compressible (transonic) flow. 7. Describe the basic steps towards non-linear supersonic flow. 8. Perform a supersonic flow experiment and calculate basic shock wave properties.	
Education Method	Lectures and self study	
Literature and Study Materials	Text book: J.D. Anderson, Fundamentals of aerodynamics, 3rd ed. (or later), McGraw-Hill Wind tunnel practical: F.F.J. Schrijer, Lab manual Supersonic Wind Tunnel Exercise (available as digital download on brightspace) Further recommended literature M.J. Zucrow, J.D. Hoffman, Gasdynamics.- vol. 1, 1976 ISBN 047198440X. J.D. Anderson, Modern compressible flow H.W. Liepman, A. Roshko, Elements of Gasdynamics, Dover Publications, 2001 P.G. Bakker, Compressible Aerodynamica, Dictaat 2-54.	
Assessment	Written exam (open questions). Bonus points can be earned upon successful completion of the high speed wind tunnel practical. The bonus points can only be used for the following exam (not for the resit).	
Set-up	Before class, students prepare for the lecture by studying the previous material + solving relevant homework problems. Lecture begins with an explanation and discussion of homework problems. New topics are then discussed, and new material evaluated. In weeks 7 to 9 a wind tunnel practical is performed. Before attending the practical, the relevant material should be studied. Directly after the practical a report has to be made and submitted. Deliverables: week 7 to 9 (depending on the practical date): High speed wind tunnel test report (to be completed directly after the practical).	

AE2135	Structural and Vibrational Analysis & Design	8
Module Manager	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	0/8/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Structural Analysis, Design, and Vibrations	
Study Goals	Being able to design and analyse a structure with respect to: - Bending - Buckling - Torsion - Shear - Cutouts - Castigliano's second theorem - Vibrations	
Education Method	Lectures, partial tests, written exam	
Assessment	Structural Analysis: two partial tests + final exam or final exam only (weighing 100%). The best option for the student counts. Vibrations: written exam	

AE2135-I	Structural Analysis & Design	5
Responsible Instructor	C.D. Rans	
Instructor	Ir. J.A. Melkert	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	Statics, Mechanics of Materials, Calculus, Differential equations	
Course Contents	The following topics will be addressed in the course a) Unsymmetrical Bending, b) Stress function solution for torsion c) Torsion of thin walled open and closed sections d) Shear of thin walled open sections (shear center) e) Multi-cell structures (bending, shear, torsion) f) Structural Idealization - Sheet-boom approximation / Multi-cell beams g) Cutouts and Reinforcement Ribs h) Energy Methods i) Buckling	
Study Goals	At the end of this course, you will be able to: (a) Calculate stresses/strains in thin walled structures using: - engineering beam theory (bending and shear) - torsion theory (closed and open sections) (b) Modify above theories in the presence of cutouts and/or taper (c) Calculate displacements using: - beam theory - energy methods (Castigliano's theorems) (d) Determine buckling loads for simple structures (beams, trusses) (e) Design such structures where you determine the geometry so that structure does not fail (thickness of skins under bending, shear and torsion; cross-sectional geometry of beams under compression)	
Education Method	Three weekly lectures, two partial tests and one final exam.	
Books	Aircraft Structures for Engineering Students, T.H.G. Megson, Sixth edition (ISBN 978-0-08-100914-7) or Seventh Edition (ISBN 978-0-12-822868-5). An ebook version of the 7th edition is available in the TU Library	
Assessment	There are two options for the assessment; Option A: The assessment has three parts: 1. Test 1 in week 2.5, duration 1.5 hrs, course material taught upto and including week 2.4 2. Test 2 in week 2.7 or 2.8, duration 1.5 hrs, course material taught upto and including week 2.7 3. Final exam, duration 3 hrs, full course material taught The value of each of the tests is 20% of the final grade. The value of the final exam is 60% of the final grade. The results of the tests are valid for one academic year only. Option B: Alternatively, students may decide to not take part in the two tests but decide to only take the final exam. In that case the final exam (duration 3 hrs) has a value of 100% of the final grade. It covers all the course material taught. The final exam in option A is identical to the final exam in option B. Only the weighing towards the final grade differs. Students do not have to indicate which grading option they want to take. Students will be awarded the better grade from each of the two options. Please note, for option A, any absence from a test will constitute a zero on that test in terms of the calculation. It is not possible to only participate in on test and have the final exam grade count towards the missed test. During the resit in Q3 there will be one final exam. The weighing towards the final grade is identical to the one for the final exam in Q2. Students can again choose option A or option B. There are no resits for the two tests.	
Permitted Materials during Tests	formula sheet (provided) and non-graphical, non-programmable standard bachelor calculator (Texas Instruments (TI) 30xb multiview)	

AE2135-II	Vibrations	3
Responsible Instructor	Dr. J. Sodja	
Instructor	V. Yaghoubi Nasrabadi	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Aerospace structures are commonly subjected to vibrations during operation. This includes harmonic loads as found in low and high-speed rotating equipment or impact loads due to collisions. For analysis and design purposes it is critical to model the response of a structure under free and forced loading conditions. This course is an introduction to vibrations in structures and covers the following topics: 1. Introduction to vibrations and modelling of structures. 2. Free vibrations of undamped, single degree-of-freedom systems 3. Free vibrations of damped systems 4. Harmonically-forced vibrations 5. Vibrations under general loading types: impulse loading, step loading, arbitrary transient loading. 6. Modal analysis: Free vibrations of undamped multiple degrees-of-freedom systems 7. Forced vibrations and damping in multiple degrees-of-freedom systems Fundamental concepts of structural vibrations are introduced in the course, such as discretization of kinematical and force connectors, choice of degrees of freedom (DOF), lumped mass approach, damped and undamped natural frequencies, resonance, critical, undercritical and overcritical damping, transfer function.	
Study Goals	General course aim: The purpose of this course is to provide an introduction to vibrations of structures and structural components. At the end of the course, students should be able to analyze basic vibration problems and be able to use this information for design purposes. Learning outcomes: Students who successfully complete the course should be able to 1. Represent an actual structure or structural component using a mass-damper-spring model. 2. Formulate and solve the equation of motion associated to the mass-damper-spring model. 3. Understand the influence of the main model parameters on the structural response. 4. Interpret the results of the simulations in terms of global cases such as resonance and under or over-damped responses.	
Education Method	Lectures, instructions and self study.	
Literature and Study Materials	Engineering Vibrations by Daniel J. Inman, ISBN (13): 9780273768449 Note: the special TU Delft edition ISBN (13): 9781787269064 has been discontinued (last batch was in 2020). The special edition contains only the chapters used in the course, hence second hand copies may still be used.	
Assessment	Written exam.	
Set-up	1. Lectures (2 hrs per week, in English) 2. Instructions (2 hrs per week, in English) for practicing the lecture topics	

AE2220	Applied Numerical Analysis and Computational Modelling	6
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	see AE2220-I and AE2220-II	
Education Period	3 4	
Start Education	3	
Exam Period	3 4 5	
Course Language	English	
Course Contents	This module introduces the foundations of numerical analysis and computational modelling. See the descriptions of AE2220-I and AE2220-II for detailed content lists.	
Study Goals	See the descriptions of AE2220-I and AE2220-II	
Education Method	Lectures, assignments.	
Assessment	In term quizzes. No final exam.	

AE2220-I	Applied Numerical Analysis	3
Responsible Instructor	Dr. R.P. Dwight	
Instructor	Dr.ir. A.H. van Zuijlen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Applied Numerical Analysis (Weeks 1-7, Dr. R.P. Dwight) A foundational course in numerical analysis. We assume good knowledge of real analysis and calculus, but no previous experience in numerical analysis. We introduce numerical methods for solving non-linear equations, ODEs, optimization problems and integrals. We analyze these in detail, with regard to consistency, accuracy, stability and convergence, theoretically and in practice. - Taylor Series and Solutions of non-linear equations. - Computer arithmetic and round-off errors. - Concepts of accuracy, consistency, stability and convergence. - Uses and limitations of numerical methods. - Solution methods for non-linear equations: - Polynomial interpolation - Statement of the interpolation conditions. - Important properties of polynomials, the Weierstrass approximation theorem. - Choice of basis: monomial, Newton, Lagrange. - Uniqueness and existence of polynomial interpolants. - Gibbs phenomena. - Advanced interpolation methods and regression - Interpolation on triangles and rectangles (patches). - Splines. - Radial basis functions and volume splines. - Least-squares regression - Numerical differentiation and integration - Finite difference formulas, truncation error, rounding error. - Quadrature based on polynomial interpolation. - Derivation of integration conditions. - Newton-Cotes and Gauss rules 2D integration based on product rules. - Numerical methods for solving ordinary differential equations (ODEs) - Properties and classification of ODEs, standard form of ODE. - Definition of a consistent ODE solver. - Linear stability analysis (von Neumann analysis), time-step criteria. - Convergence analysis. - Single and multistep methods: - Numerical optimization - Define the unconstrained optimization problem - Local and global optima, design space, cost function, derivatives. - Quadratic forms as an example problem. - Common optimisation methods	
Study Goals	At the end of this part of the course, the student should be familiar with: - The Taylor Series and Solutions of non-linear equations - Polynomial interpolation - Advanced interpolation methods and regression - Numerical differentiation and integration - Numerical methods for solving ordinary differential equations (ODEs) - Numerical optimization	
Education Method	Lecturing, homework assignments, programming notebooks, self study.	
Literature and Study Materials	Complete lecture notes will be provided for the course. They are available as a PDF on Brightspace, and in a printed+bound version which can be ordered at the faculty. A text-book is not necessary, but if desired our recommendation is: R.L Burden and J.D. Faires "Numerical Analysis" 9th edition (paperback) Brooks/Cole Cengage (2011) Which is stocked by the VSV.	
Assessment	The course will be assessed either: (a) with a single written FINAL EXAM of 3 hours during the Q3 exam weeks, or (b) 3 partial exams during the quarter. Which form will be chosen by the lecturer, and depends on scheduling. All will be a written, multiple-choice exams. Bonus points can be obtained by solving Python programming exercises. A resit in the exam period of Q4, consists of a single written multiple-choice exam on paper. Bonus points do not apply.	
Set-up	Lectures (2 hours/week): Pre-recorded video lectures to be followed by self-study. Inverted classroom (2 hours/week): Elaboration on the video, answering questions on material, doing example problems from the homeworks. Self-study (4 hours/week): Weekly homework assignments with practice analytical problems and (Python) programming.	

AE2220-II	Computational Modelling	3
Responsible Instructor	Dr. S.J. Hulshoff	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Complex physical phenomena, from flows to structural deformations, are often modelled using partial differential equations (PDEs). This course provides an introduction to the numerical solution of PDEs. We begin by describing the general principles of computational modelling, and investigate the properties of PDEs derived from common physical problems. We then describe two popular discretisation approaches, the finite-difference and finite-element methods, along with techniques used for their analysis and practical application. In the last part of the course, we consider time-march methods for unsteady problems, and basic iterative techniques for the solution of large algebraic systems. 1. Introduction to computational modelling - Errors in the numerical representation of physical phenomena - Requirements for discretisation methods 2. Classification and boundary conditions - A review of model PDEs - Basic classification of PDEs - Definition of characteristics, Elliptic, hyperbolic and parabolic PDEs - Dirichlet and Neumann boundary conditions, well-posed problems 3. Finite-difference methods - Modified equation, Taylor table - Upwinding and artificial dissipation - Generalised transformation - Fourier analysis 4. Verification - Method of manufactured solutions - Code and solution verification, Richardson extrapolation. 5. Spectral and finite-element methods - Method of weighted residuals - Element and global assembly - Application of boundary conditions - Unsteady problems 6. Analysis of time-march methods - Accuracy of transient computations - Systems of ODEs, semi-discrete eigenvalues - Wave space, relation to Fourier analysis - Fully-discrete eigenvalues, stability, stiffness 7. Iterative solution of algebraic systems - Direct solution methods versus iterative solution methods - Jacobi and Gauss-Seidel techniques - Convergence rates, stopping criteria	
Study Goals	At the end of this part of the course, the student should be able to: 1. Select appropriate numerical discretisation techniques for a given PDE 2. Formulate physically consistent numerical boundary treatments. 3. Analyse numerical discretisations in terms of their accuracy and stability. 4. Select a time-march method well-suited to a given problem. 5. Understand the design and behaviour of iterative solution techniques. 6. Verify the results of numerical simulations	
Education Method	Lectures, work sessions, tutorials and self study	
Literature and Study Materials	Complete lecture notes, citing a collection of optional references per topic.	
Assessment	Based on results from an exam and three optional work sessions, or a resit exam.	
Set-up	Study Material: - Lectures: Presentation and discussion of concepts and examples - Notes, slides, and practice problems released via Brightspace - Work Sessions (optional): Three sessions consisting of a preparation followed by an in-class group activity. Both of these are assessed. - Tutorials (optional): Three sessions consisting of a discussion of the practice problems students performed weakly on, followed by an opportunity for individual questions and answers. Students must complete the practice problems before attending the tutorial. Marking: The mark will be determined from the best of the three options below: 1. 70% (Final exam) + 30% (Worksession average) 2. 100% (Final exam) 3. 100% (Resit)	

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AE2224	Test, Analysis and Simulation	8
Contact Hours / Week x/x/x/x	See AE2224-I and AE2224-II	
Education Period	3 4	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course deals with analysis and interpretation of data from aerospace engineering problems	
Study Goals	During this course you will learn to: 1. Critically assess scientific literature in the field of aerospace engineering 2. Formulate appropriate hypotheses 3. Analyse and interpret scientific data, using modern data analysis techniques 4. Write a research report and present data to others	
Education Method	The teaching method is a combination of lectures and scheduled project sessions. Attendance is compulsory. Please check the course descriptions for more details on absence rules.	
Assessment	Final reports and oral exam. Check the course description for more details	

AE2224-I	Test, Analysis & Simulation	5
Responsible Instructor	Dr.ir. E.J.O. Schrama	
Responsible Instructor	J. Sun	
Responsible Instructor	Dr. B. Chen	
Instructor	G. Kelesidis	
Contact Hours / Week x/x/x/x	0/0/8/8	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Students work in groups of ~10 on projects that deal with the topics of test, analysis & simulation. These projects are defined and supervised by academic staff from the Aerospace Faculty of the Delft University of Technology. The work to be carried out by the group consists of at least the following activities: 1. Analysis of the data; 2. Verification of data analysis results through comparison with theory, simulations, and literature; 3. Interpretation of the results, resulting in conclusions with regards to the research question and recommendations; 4. Documentation of the research question, test, simulation and analysis results, and the conclusions in a scientific report; 5. Write a self-reflection. The student should reflect both on his/her technical input and functioning in the research group; 6. Peer review of reports as written by students from other groups.	
Study Goals	Collecting the data is not a part of the project. The thematic projects in the Aerospace Engineering Bachelor Program aim to provide learning experiences for students that enable them to integrate the (theoretical) content of thematic courses in a practical, active setting. They aim to simulate a professional environment in which students act within a professional role, on real-life problems which result in a professional output product, such as a technical report, a poster, an abstract, a scientific presentation, or a scientific paper. The projects are mandatory elements of the program. Each semester contains one thematic project. The theme of this 4th semester project is data analysis and evaluation. The outcome of the project work is to be documented in a scientific report. The aim of the project is that at the end of the project, the student will be able: 1. To analyse an experimental set-up or simulation model and assess its performance with regards to answering the posed research question; 2. To carry out an analysis of experimental and/or model results; 3. To draw conclusions with the aim to answer a research question; 4. To write a research report and present the research to others; 5. To function as a member of a research team in a research environment; 6. To be able to critically reflect on one's own research as well as others using a peer format.	
Education Method	Project education	
Prerequisites	The entrance requirements are: 45 ECTs of the first year of the programme, a positive BSA and the two first year projects AE1111-I and AE1222-I should be completed. The number of credits includes AE1111-I and AE1222-I and is based on the results up to and including the first examination period (October).	
Assessment	The end mark for the project is comprised of the following components: 1. Scientific report; 2. The attitude of the student during the project; 3. The oral exam at the end of the project; 4. A grade for the peer review. Items 1, 2, 3, and 4 are weighted equally. The self-reflection report will be accounted for in the grade for item 2. Item 1 is a group grade, whereas items 2, 3, and 4 are individual grades. The grade will be given by the tutor of the group and holds for the whole project grade. The "Scientific Writing" course is assessed with a pass/fail grade on the quality of the scientific report. Students will only be awarded a passing grade (6.0 or higher) for the project if: 1. The whole project has been completed and all compensatory assignments have been completed successfully and, 2. All rules regarding absence are met and, 3. No more than one grade is lower than 6.0 and, 4. All grades are 5.0 or higher. A pass grade is obtained for the "Scientific Writing" course. The grade for the project will be rounded to 1 decimal. Not meeting the above requirements results in a grade 1.0.	
Set-up	The grade for the project is combined with the grade for AE2224-II for determining the AE2224 module grade. Throughout the semester, students are expected to work on the project during the scheduled hours. In addition the student should participate in the coaching sessions on the topic of Scientific Writing. Attending the scheduled project sessions is obligatory and will be registered. The following rules regarding absence hold: 1. Students are allowed to miss a maximum of 2 project sessions during the 1st period of the semester, 1 project session during the 2nd period of the semester for the project itself and only for exceptional circumstances (illness, family circumstances or relevant extenuating circumstances). 2. If a project has a supporting course the student is only allowed to be absent for one session of this course. This session is not counted as a missed project session as mentioned under 1. 3. Missing time (either by being late or leaving at any time) during project sessions can result in the student being registered as having missed sessions. 4. Missed sessions must be compensated according to the rules stated below 5. The student must attend the first week of a project. Not being present in the first week of the project results in exclusion from participation in the project in that academic year 6. A student is not allowed to miss the last two sessions of the 1st period together with the first session of the 2nd period of a semester. 7. Absence for more than the allowed number of sessions or failure to make up for missed hours will mean that the student cannot obtain a pass grade for the project. 8. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counsellors for guidance. For compensating absence, the following applies:	

1. The student is responsible for fulfilling the compensatory assignment, at a time or within a time limit set by the responsible lecturer (this also applies to supporting courses).
2. The quality of the compensatory assignment is assessed by the responsible lecturer. If the student does not complete the assignment within the allocated time and/or with sufficient quality, they will fail the project.
3. Compensatory assignments cannot be amended or redone if deemed unsatisfactory upon assessment by the responsible lecturer.
4. No opportunities are offered outside the duration of the project for making up for missed sessions.
5. Students who are of the opinion that there are relevant extenuating circumstances can turn to the academic counsellors for guidance.

Deliverables:

period 3:
week 4: literature review and research plan
week 7: self reflection

period 4:
week 4: Submit draft of scientific report for the peer review process
week 6: Submit peer review and individual abstract
week 8: Submit final version of scientific report to Brightspace, tutor and scientific writing teacher

AE2224-II	<i>Artificial Intelligence for Aerospace Engineering</i>	3
Responsible Instructor	Dr. D. Zarouchas	
Responsible Instructor	Dr. B. Giovanardi	
Instructor	Dr. G.C.H.E. de Croon	
Instructor	N. Eleftheroglou	
Instructor	N.D. Eskue	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	-AE1110 Introduction to Aerospace Engineering; -W11403LR Linear Algebra; -AE1205 Programming & Scientific Computing in Python for AE1; -W12180-I Differential Equations; -W12180-II Probability and Statistics.	
Course Contents	Each week will consist of one 2-hour frontal lecture and one 2-hour coding laboratory. During the laboratory sessions, the instructors will guide the students in the hands-on application of what learnt in class to real world problems in Aerospace Engineering. This course will cover the following topics in 6 weeks: - Regression - Data Reduction - Classification - Clustering - Artificial Neural Networks In the last week of the course, field experts will be invited as external lecturers to discuss relevant present-day topics in Artificial Intelligence for Aerospace Engineering, including but not limited to: - Ethical Issues in Artificial Intelligence; - Machine Learning and Certification; - Sustainability and Machine Learning.	
Study Goals	At the end of the course, students will be able to: - Describe the most-commonly used Machine Learning models and their scope of applicability; - Match the appropriate class of models to a given Aerospace Engineering application; - Understand the range of applicability and the limitations of the models studied; - Apply the models studied using well-established open-source Python packages; - Understand and use concepts and jargon borrowed from the Artificial Intelligence community in the context of problems in Aerospace Engineering; - Apply conventional Artificial Intelligence algorithms to real-life problems in Aerospace Engineering.	
Education Method	Lectures and coding laboratories (Python).	
Assessment	Final written (coding) exam on Weblab with multiple choice questions, open questions, and coding problems.	
Tags	Algebra Algorithmics Artificial intelligence Computer Engineering Databases Ethics Image processing Mathematics Mechanics Numeric Methods Practicals Programming	

AE2230	Flight & Orbital Mechanics and Propulsion	8
Responsible Instructor	A. Cervone	
Contact Hours / Week x/x/x/x	See AE2230-I and AE2230-II	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The module AE2230 consists of two elements; a course on Flight and Orbital Mechanics (AE2230-I) and a course on Power and Propulsion (AE2230-II). A detailed description of the contents and learning goals of these two courses can be found in the study guide, as well as information regarding practical matters (book, education method, etc.).	
Study Goals	Study goals are described in the courses AE2230-I and AE2230-II	
Education Method	The educational methods are described in the courses AE2230-I and AE2230-II	
Assessment	The methods for assessment are described in the courses AE2230-I and AE2230-II	

AE2230-I	Flight & Orbital Mechanics	4
Responsible Instructor	Dr. C. Varriale	
Instructor	J.G. De Teixeira da Encarnacao	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	1. Turning performance (three dimensional equations of motion, coordinate systems, Euler angles, transformation matrices) 2. Airfield performance (take-off and landing) 3. Unsteady climb and descent (including minimum time to climb problem) 4. Cruise flight and transport performance 5. Effect of a constant wind field on selected performance metrics 6. Equations of motion applied to various phases of space flight 7. Launch, Vertical flight, delta-V budget, burn out height, staging 8. Perturbations on satellite orbits 9. Spacecraft maneuvers and transfers 10. Eclipse 11. Interplanetary flight	
Study Goals	1. Integrate fundamental disciplines (aero, power and propulsion, mechanics..) to describe the kinematics of aerospace vehicles satisfying real world constraints 2. Derive equations of motion for elementary flight and mission phases (climb, turn, cruise, take-off, launch, orbit) 3. Derive analytical expressions for optimal performance (steepest turn, Breguet Range, patched conics, J2, maneuvers) 4. Determine pros/cons of multi-stage launchers. 5. Assess sun lighting conditions on a satellite. 6. Determine the influence of wind (gradient) on aircraft motion and performance. 7. Develop the theory to describe an interplanetary trajectory as a succession of two-body problems, and apply this concept to real missions.	
Education Method	Lecturing and self study.	
Literature and Study Materials	G.J.J. Ruijgrok, "Elements of Airplane Performance," 2nd Edition (not mandatory, available at https://repository.tudelft.nl/islandora/object/uuid:9ff3c7e0-e221-4d8f-bf9e-e144d3cacdc1?collection=research) Asselin, Introduction to Aircraft Performance (not mandatory) Gudmundsson, General Aviation Aircraft Design: Applied Methods and Procedure (not mandatory) Fortescue, P., J. Stark and G. Swinerd: Spacecraft Systems Engineering (3rd edition), Wiley & Sons, 2003. (For further reading, not mandatory) Wertz, J.R., and W.J. Larson: Space Mission Analysis and Design (3rd edition), Space Technology Library/Microcosm/Kluwer, 1999. (For further reading, not mandatory).	
Assessment	Students do a digital exam in which they get a mix of open and multiple-choice questions. They are tested with respect to knowledge and insight and have to make calculations and analytical derivations.	
Set-up	Students will follow an active lecture, in a large group. The lectures will consist of a mixture of introduction of the theory, derivations, example questions and interaction with students.	

AE2230-II	Propulsion & Power	4
Responsible Instructor	A. Cervone	
Instructor	Dr. I. Langella	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	The AE2230 course is required for several MSc courses listed below 1. AE4238 - Aero Engine Technology 2. AE4237 - Heat Transfer Problems in Gas Turbines 3. WB4421 - Gas Turbine Simulation 4. AE4S01 - Thermal Rocket Propulsion.	
Expected prior knowledge	It is expected that the students should have a good knowledge about: 1) the fundamentals of physics as taught in the module AE1240 (Physics) 2) the importance of the propulsion and power subsystem for the aerospace vehicle as taught in AE1222-II - Aerospace Design and Systems Engineering Elements	
Course Contents	Air Breathing Propulsion Aircraft Propulsion:- thermodynamics and cycle calculations, Working of a Gas Turbine Engine, turbo machinery, combustion, design point calculations. Non-Air Breathing Propulsion Rocket systems: Ideal rocket motor theory, cold gas rocket systems, chemical rocket systems. Electrical Power Systems Electrical Generators, Photo voltaics, Batteries, Fuel Cells, Power Management and Distribution.	
Study Goals	At the end of this course, the student will be able to: 1. Understand the basic principles of thrust and power producing mechanisms for aerospace vehicles. 2. Perform basic sizing of thrust and electric power generation systems suitable for aerospace vehicles. 3. Describe the various components of a gas turbine engine, their working principle and performance. 4. List/describe/explain -The main thrust and electrical power generation options available -(Main) components that make up the propulsion and electrical power generation system and their function -Current limits to thrust/power generation. 5. Apply control volume analysis and integral momentum equation to estimate thrust produced 6. Apply physics to predict electric power generated by solar photo-voltaics, batteries, electrical generator, etc. 7. Develop system model from schematic system description. 9. Size electrical power system for a given mission. 10. Select appropriate propulsion/power system from basic types depending on system requirements. 11. Assess effect of changes in design/operating parameters on system performance.	
Education Method	Lectures, videos, assignments and self study. There are also practice questions and exercises on Brightspace. Bonus assignments will be given to the students. However the bonus points will be accounted in the final grade only if the student earns a minimum of 5 grade points in the final exam.	
Literature and Study Materials	- Lecture slides (available on Brightspace). Supporting study material on Brightspace: - Readers. - Videos. - Sample exams and their solutions. - Practice Mobius questions.	
Books	Books are NOT mandatory for this course. However for more information, the students are referred to the following books: 1. Gas Turbine Theory : By Henry Cohen, G. F. C. Rogers, H. I. H. Saravanamuttoo, Publisher: Addison Wesley Longman 2. Spacecraft Systems Engineering, 3rd ed. : By Peter Fortescue, John Stark and Graham Swinerd, Publisher: John Wiley & Sons.	
Assessment	Closed book exam with open and multiple choice questions. The exam is mostly digital, possibly with a limited number of questions to be answered on paper. A formula sheet is given during the exam. Assignments will be given during the course which can provide bonus points to the students. Details on the assignments are given on Brightspace and during the lectures. Students can earn a maximum of 1 bonus point (0.5 for the air-breathing propulsion part and 0.5 for the Electric Power System+Rocket Propulsion part). Only students who score 5.0 or more grade points in the final exam are eligible for the bonus points.	
Permitted Materials during Tests	1. Normal calculator. Graphical calculators are also allowed, under the condition that the memory is cleared before the exam starts. 2. Formula sheet provided by the instructors during the exam.	
Set-up	The course is offered in the BSc second year (second semester). Students attend classes (28 lecture hours), but some learning goals are linked to videos which will be made available on Brightspace. Some assignments will be given which can be completed by the students to earn bonus points. Students can also do	

the practice exercises/assignments posted on Brightspace.

The final assessment is based on an exam (partly written, partly electronic) + the bonus points earned by the student.

AE2235	Aerospace Signals, Systems & Control	7
Module Manager	Dr.ir. M.M. van Paassen	
Contact Hours / Week x/x/x/x	See AE2235-I and AE2235-II	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This module contains two courses, AE2235-I, Aerospace Systems and Control Theory and AE2235-II, Instrumentation and Signals. For a detailed overview of the contents, see these individual courses.	
Study Goals	At the end of this module, the student will know how to or be able to: - do signal acquisition, processing and filtering - apply frequency domain techniques to signal conditioning and processing - know building blocks and electronic components - apply signal modulation, produce a signal link budget - formulate linear system models, combine models, create control structures - determine dynamic properties of linear systems, identify desirable properties - evaluate stability properties of closed and open loop systems - tune control systems with classical tuning methods In summary, the student will have the tools and knowledge to work with linear dynamical systems and their input and output signals. In AE2235-I, the focus will be on systems modelling, in AE2235-II the focus will be on signal analysis.	
Education Method	For a more detailed description of the study goals, consult the descriptions for AE2235-I and AE2235-II For AE2235-I: - 14 oral lectures, 2 hours each - 7 e-learning modules, to be completed with Python with python-control installed, or with Matlab For AE2235-II: - 12 oral lectures, 2 hours each - a studio classroom session, which will include hands-on experimentation with hardware sensors and data acquisition equipment. At the session the data is analysed with Python or Matlab.	
Assessment	AE2235-I: Computer based exam, 2 hours. Sign up for the exam is through Osiris. During the exam, students have access to Matlab with the control toolbox and to Python and the python-control module. See BrightSpace sites for ae2235-I for information on the available editors in the exam environment, if you used other editors, plan on practicing a bit with editors in the exam environment. AE2235-II: Written exam, combination of open and multiple choice questions.	

AE2235-I	Aerospace Systems & Control Theory	4
Responsible Instructor	Dr.ir. C.C. de Visser	
Instructor	Dr.ir. M.M. van Paassen	
Instructor	S. Theodoulis	
Contact Hours / Week x/x/x/x	0/0/0/4 of which 7 E-lectures	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course teaches the elements of handling dynamics in systems encountered in Aerospace Engineering. The goal is to provide students with all the theory and tools necessary to enable them to design, implement, and operate control systems for aerospace systems. An important part of the course is the actual programming (in Python/Matlab) of control systems for a wide variety of aircraft, UAVs, and satellites. For this, the student is introduced to the concepts of dynamical systems; inputs, outputs and system boundary and environment. Students learn to connect different system descriptions; state-space models, transfer functions, differential equations and frequency response descriptions. Students learn block diagram descriptions and block diagram manipulation. Desirable and undesirable properties of control systems are distinguished, the choice between elementary controller types to match a specific controlled system is discussed. Tuning methods for the controller parameters include the root-locus method, and frequency design methods (Bode). The course uses examples commonly found in aerospace engineering, both dynamic models for whole systems (aircraft or satellites) and models for components, such as a landing gear or control surface with a hydraulic servo.	
Study Goals	At the end of this course, the student will be able to: 1. Given a schematic of a physical system and desired behavior, design a simple (single-loop) controller. 2. Formulate linear dynamical system models from block diagrams or schematic mechanical system descriptions. 3. Identify dynamical properties across domains time, frequency, transfer function, A-matrix eigenvalues. 4. Evaluate stability of open and closed loop systems (in different domains: time, frequency, on the basis of pole locations) 5. Calculate and judge responses to input signals in time and frequency domain. 6. Select appropriate controller from basic types (P, PI, PD, PID, lag-lead, lead-lag) depending on system properties and requirements. 7. Tune controllers with Bode, Nyquist and Root-locus methods (Evans) with a computer. All the above tasks will be done using a Computer Aided Control System Design (CACSD) tool, in this case either Matlab or Python.	
Education Method	Lecturing (12 oral lectures), complemented by (7) on-line E-learning modules. For support with the e-learning modules, both online and physical instruction sessions are scheduled.	
Literature and Study Materials	Norman S. Nise, Control Systems Engineering (5th edition or 6th edition), Wiley & Sons (Optional) Presentation slides and additional material on Brightspace Online exercise material - students are advised to add their own notes to that material and download/print (or convert to pdf) this material, to create a personalised study text.	
Assessment	Computer exam, 2 hours. Combination of open and multiple choice questions.	
Remarks	The course forms a module with ae2235-II Instrumentation and Signals.	
Set-up	The course is given as a combination of lectures and on-line exercises (e-learning). In the lectures, new topics are presented. In the e-learning exercises, the students work with a CACSD tool (Matlab or Python), to get hands-on experience with the material presented in the lectures. Be aware that this is not the same as an exercise accompanying a lecture series; there is significantly more interaction. An e-learning lecture can also include calculations whose results are explained and used in the following lecture. Week arrangement (e-lectures indicated with <e>): 1 - Introduction, open and closed-loop control, input, output, system concepts, control error 2<e> - Matlab skills for control theory, experimenting with a simple control system 3 - Transfer functions, mechanical systems 4<e> - Transfer functions in Matlab, combining transfer functions, response calculations, control systems requirements 5 - State-space systems, aircraft equations of motion as state-space 6<e> - State-space in matlab, response calculation with state-space 7 - Transient and steady-state responses, system type, position, velocity, acceleration error, basic controller types 8<e> - Entering more complex models in Matlab. Combining transfer functions, state-space systems and block diagrams. 9 - Root-locus tuning 10<e> - Root-locus tuning in Matlab, using aircraft autopilot modes. 11 - Frequency response, Bode diagram, non-minimum phase systems 12<e> - Exercise in frequency response, using a flexible satellite. Notch filters. 13 - Stability in the frequency domain. Polar plot, Nyquist contour, Nyquist diagram. 14<e> - Combination of root-locus and frequency domain techniques, applied to a model of an unstable rocket. Deliverables: week 2: E-lecture 1 finished week 3: E-lecture 2 finished week 4: E-lecture 3 finished	

week 5: E-lecture 4 finished
week 6: E-lecture 5 finished
week 7: E-lecture 6 finished

AE2235-II	Signal Analysis and Telecommunication Principles	3
Responsible Instructor	Prof.dr.ir. M. Mulder	
Instructor	Dr.ir. D.M. Pool	
Practical Coordinator	F.N. Postema	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Week 1, Lectures 1-2 Introduction; Basic principles of signal acquisition, conditioning, modulation and transfer; Definitions (continuous time, discrete time, periodic/a-periodic); Basic signal shapes (unit pulse, step, ramp, sinusoid); Refresh complex algebra, Euler's theorem; Introduce Unit impulse function (Dirac, sifting property) Week 1, Lectures 3-4 Signal decomposition; Fourier Series (real and complex exponential versions); Sinc function; Examples. Week 2, Lectures 5-6 Fourier Transform, Basic Transforms, Duality, Transform pairs, Properties of FT, Convolution; Examples. Week 2, Lectures 7-8 Relation Fourier Transform and Fourier Series; Examples Energy and Power, Parseval's Theorem; Definition of Energy and Power Spectral Density; Examples. Week 3, Lectures 9-10 Introduction to linear time-invariant systems (LTI); Impulse response function, Transfer function; Examples. Week 3, Lectures 11-12 Fourier Analysis, Frequency-response function; Filtering, filtering properties (bandwidth, rise time); Examples. Week 4, Lectures 13-14 Sampling; A/D and D/A conversion; Impulse-train sampling and signal reconstruction; Nyquist sampling theorem, aliasing; Examples. Week 4, Lectures 15-16 Introduction to Discrete Fourier Transform (DFT) and Fast-Fourier Transform (FFT); Examples Week 5, Studio Classroom Session Sampling, aliasing, windowing, leakage. Basic signal conditioning; Low-pass Filter. Using data from actual aerospace sensors (accelerometer, pitot tube, rate sensor). Week 6, Lectures 17-18 Basic principles of transferring information (communication); modulation (digital system, binary signaling); On-Off-Keying, and Binary Phase Shift Keying; Time and frequency representation. Week 6, Lectures 19-20 Effects of noise; Additive White Gaussian Noise (AWGN); thermal noise, noise temperature, noise density, effective noise bandwidth; Signal-to-Noise-Ratio (SNR); Signal detection. Week 7, Lectures 21-22 Optimal signal detection (false alarm and missed detection probabilities); range estimation for navigation and surveillance; examples in aerospace; signal bandwidth versus bit-rate (communication) link/channel capacity, and bit energy to noise density ratio E_b/N_0. Signal bandwidth versus ranging accuracy (navigation), chip/pulse duration, and carrier to noise density ratio C/N_0. Week 7, Lectures 23-24 Design calculations for telecommunications sub-system in aerospace; basic radio (wireless) signal link budget (and radar equation); Emitted Isotropic Radiation Power (EIRP), free space loss, and antenna gain. Examples: satellite-Earth link, aircraft-tower link, and radar two-way sensing; quick review of transmitter and receiver building blocks.	
Study Goals	At the end of this course, the student will be able to: 1. How to acquire and condition a signal (from a transducer) suitable for further processing? 2. How to transfer a signal from A to B? 3. Comprehend signal representation/decomposition in time domain and frequency domain 4. Design a filter to condition signal (including trade-offs in performance) 5. Design a sampling scheme 6. Apply filtering and sampling to an actual case, and evaluate the result 7. Comprehend limitations and constraints (quantization, signal-to-noise) 8. Know building block, electronic components, and comprehend main functions 9. Comprehend signal modulation techniques 10. Apply & implement modulation to a (simplified) real signal 11. Comprehend signal detection techniques 12. Produce a signal link budget	
Education Method	Lecturing and self study. A studio classroom session is planned to demonstrate the concepts of signal conditioning, filtering, and sampling.	
Literature and Study Materials	The material consists of: [1] Selected chapters from the book "Signals and Systems - Continuous and Discrete", fourth edition, by Ziemer, Tranter and Fannin. Pearson International Edition. [2] The lecture slides, which will be put on the Brightspace. [3] Additional hand-outs, which will be put on the Brightspace. [4] Reader for course AE2215 "Instrumentation and Signals - signal modulation and detection", by C.C.J.M. Tiberius, edition May 2015 (free available on Brightspace).	
Assessment	Written exam (open and multiple choice questions). In the Studio Classroom session groups of 2 students will submit an answer sheet. When the answers in this sheet are correct, and the grade for the written exam is 6.0 or higher, students can gain a full bonus-point (+1.0) for the examination. The bonus point will be valid only in the academic year where the studio classroom sessions have been conducted, i.e., for the exam and the resit.	

Permitted Materials during Tests	Formula sheet and normal, non-programmable calculator.
Set-up	On a typical day, a student participates in the lecture, where the lecturer explains the (mathematical) background of signal acquisition, conditioning and transfer problems and performs the basic calculations. Students are advised to do the homework assignments (Ziemer exercises and old exams). On another day, some weeks later, the student participates in a group of 2 students in a studio classroom session, complementing the 14 more traditional lectures. The groups are working on experimental data (coming from real aerospace sensors) and are required to do some elementary manipulations that correspond with the progress made during the active lecture. The classroom sessions focus on two of the most important themes: Filtering, and Sampling. PYTHON is the platform of choice, and example py-files and data files will be made available to the students. The first two hours are used to let groups work on the basics of FFT on data; the second two hours are used to let groups work on their experimental set-up, measuring real-life data and working on that. The practical aims at including data coming from real aerospace sensors, like accelerometers, rate gyros and other sensors. Deliverables: week 5: Day depends on group A-D, Filled-in answer sheet from the studio classroom

WI2180LR	Differential Equations & Probability and Statistics	8
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WI2180LR-I	Differential equations	4
Responsible Instructor	Dr. A.B.T. Barbaro	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Many Aerospace Engineering courses depend and rely on the concepts and techniques covered in this course on Differential Equations, as do courses on Numerical analysis and Partial differential equations.	
Course Contents	First order differential equations. Existence and uniqueness of solutions: initial value problems. Linear versus nonlinear equations. Integrating factor method and Separations of Variables. Second order linear differential equations. The superposition principle, general solution, linear (in)dependence, fundamental set of solutions, Wronskian, Abel's Theorem. Order reduction, variation of parameters. The Laplace transform; linear differential equations with discontinuous forcing. Systems of differential equations, both linear and locally linear. Phase plane analysis. Linear partial differential equations. Method of separation of variables, including Fourier series. Boundary value problems. Selected applications, including diffusion, wave and Laplace equations.	
Study Goals	Identify different types of differential equations and the methods used to solve each type. Perform (Forward) Eulers Method on Initial Value Problems. Use the characteristic equation to solve second-order linear equations. Use the Laplace Transform to solve Initial Value Problems, particularly ones with discontinuous or impulsive forcing functions. Solve systems of first-order linear differential equations. Analyze systems of nonlinear differential equations, including their phase planes. Solve linear PDEs (e.g. Heat, Wave, and Laplace Equations) using Separation of Variables.	
Education Method	Lecture 6/0/0/0	
Literature and Study Materials	W.E. Boyce, R.C.DiPrima, Elementary differential equations and boundary value problems, Wiley, New York; Materials as offered through the Brightspace pages of the course.	
Assessment	Three hour written exam, closed book, with no electronic aids allowed. Please make sure to enroll for course in Brightspace to receive all relevant communications and updates.	
Co-Instructor	Dr. J.L.A. Dubbeldam	

WI2180LR-II	Probability and Statistics	4
Responsible Instructor	N. de Kleijn	
Contact Hours / Week x/x/x/x	6/0/0/0 4 hours of lectures during the first 6 weeks and 2 hours of working classes during week 2-7	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	This course covers the basics of probability theory and statistics. Topics in probability include discrete and continuous random variables, probability distributions, conditional probability and stochastic independence, joint distributions, sums and functions of random variables, expectation, variance and covariance, the law of large numbers, and the central limit theorem. Topics in statistics include random sample and sample distributions, graphical summaries, estimation and (un)biased estimators, maximum likelihood estimation and least square estimation, confidence intervals, hypothesis testing and type I and II errors, the t-test and statistical learning.	
Study Goals	At the end of the course, the students will be able to: - Relate random variables to statistical experiments and select statistical models suitable for describing real-life situations; - Perform computations with random variables and apply the law of large numbers and the central limit theorem; - Identify and use numerical values that summarize the information contained in random variables; - Analyze data by means of graphical and numerical summaries; - Define and evaluate statistical procedures for conducting statistical inference.	
Education Method	Lectures (4 h/w, in week 1 to 7). Education will take place in a blended way: the student watches a prelecture video, makes some exercises related to this, participates in the lecture (which consist of lecturing by the teacher and making exercises) and then makes some more exercises at home and during exercise classes. This cycle is repeated for every lecture. Additionally there are a handful of computer assignments to work practically with data.	
Books	A Modern Introduction to Probability and Statistics Understanding Why and How Springer Texts in Statistics, Dekking, F.M., Kraaikamp, C., Lopuhaä, H.P., Meester, L.E. 2005, XVI, 488 p. 120 illus., Hardcover, ISBN: 978-1-85233-896-1	
Assessment	Two exams: midterm in week 5 about first part of the course (probability theory) and a final exam in week 9 about the second part of the course (statistics). The final grade will be the average of the midterm and final exam. Students can obtain a maximum of 1 bonus point through the computer assignments. Bonus points are only awarded if the average grade is greater than a 5.0.	
Tags	Stochastics	
Co-Instructor	Prof.dr. A.J. Cabo	

Year	2024/2025
Organization	Aerospace Engineering
Education	Bachelor Aerospace Engineering

LR BSc 3e jaar 2024

Year	2024/2025
Organization	Aerospace Engineering
Education	Bachelor Aerospace Engineering

LR BSc 3e jaar Major

AE3200	Design Synthesis	15
Responsible Instructor	Ir. J.A. Melkert	
Instructor	Dr.ir. O.K. Bergsma	
Instructor	Dr. M.D. Pavel	
Instructor	Dr.ir. C.J. Simao Ferreira	
Instructor	Dr.ir. E. Mooij	
Instructor	Dr. J. Guo	
Contact Hours / Week x/x/x/x	0/40/0/0 0/0/0/40, this course is offered twice per year	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The entry requirements are: Spring DSE (fourth period) First and second year fully completed Registered on time for the exercise Third year BSc-student or older Selection for the Spring DSE takes place on the basis of results up to and including the second examination period (January). Fall DSE (second period) First year and second year fully completed Registered on time for the exercise Fourth year BSc- student or older Selection for the Fall DSE takes place on the basis of results up to and including the summer resit exam period (July). The registration for the Spring DSE will open in week 2.5 and close by the end of week 3.1. The registration for the Fall DSE will open in week 4.5 and close by the end of week 1.1. There will be only one message posted on Brightspace when registering in Osiris is possible. It is the responsibility of the student to watch this closely. Failing to register on time will automatically exclude a student from participating in the DSE for that period. Please contact the academic counsellor in case of special mitigating personal circumstances.	
Course Contents	The Design Synthesis Exercise (DSE) students are required to make a design in the field of aircraft, spacecraft or wind turbines. In the design project the student is provided with an opportunity to obtain "design experience". This means that the student goes through the complete design process, from drawing up a set of detailed (mission) requirements, concept analysis and design, concept selection to the presentation of the final design, in a structured and iterative manner. He/she will experience the difficulty of making well-motivated design choices, thereby taking into account (sometimes conflicting) demands, etcetera. He/she will also experience that design iterations are necessary to tune non-optimal design decisions to meet the specifications drawn up at the start of the exercise. Some examples of aerospace vehicles designed in the past include: High-altitude VTOL rescue Thin Haul Transport and Air Taxi Water Bomber UAVs Aircraft Maintenance Drone Returning to Saturn - Characterizing the Icy Moons and Rings Demonstrator for Airbreathing Electrostatic Propulsion, Earth Observation Satellite A Personal Air Mobility Vehicle as Maritime Pilot Shuttle A stratospheric balloon as operational and test platform for Earth observation. eVTOL emergency aircraft Develop of a secondary rotor wind turbine Low-cost system for stratospheric aerosol injection Adaptive Regional Airliner The credits that can be gained by successfully completing the exercise are 15 EC. Students who have suggestions for design topics can bring them forward to the responsible instructor. The idea must be within the topics of aircraft, spacecraft or wind turbine design and concern the design of a complete object. Parts of these objects, mission designs without hardware components or procedures are not considered suitable for the DSE. If the idea seems feasible, a suitable principal tutor will be sought. When a suitable principle tutor can indeed be found the text of the assignment can be written and put forward to the external quality control committee. Together with this principal tutor the student can draft the assignment. The assignment will have to pass the regular quality control procedure. If successful, the assignment will be offered to all the participants of the DSE in either the Spring of the Fall DSE. The topic of the design is limited to aircraft, spacecraft or wind turbines. Participation of those that have helped with the set-up of the design assignment is limited to two students per design team.	
Study Goals	In the design project, the students must demonstrate that they have the basic knowledge and skills necessary to accomplish a successful "paper" design of an aerospace system. By completing the project, the student will demonstrate: - Technical competence and ability to apply knowledge - Design competence (Perform conceptual design of an aircraft or spacecraft system, integrate life-cycle and sustainability issues in the design) - Effective communications (plan, prepare, deliver and assess meetings, oral presentations and written reports). - Professional attitude. - Work in multi-disciplinary teams - Manage their work - Perform peer and self reviews - Understand contemporary & societal issues in their work - Exhibit life long learning attitudes and abilities	
Education Method	Project (full working week, 9 AM - 6.00 PM, every day). This timing is based on an 8 hr working day. A typical day would look	

Literature and Study Materials	like this: 8 hours of working + 2 * 15 minute tea/coffee break + 30 minute lunch break.
	The working times can be adjusted a bit (+/- 30 minutes) in consultation with the principal tutor.
Assessment	Will be communicated prior to the start of the exercise. This will vary with the assignments.
	The work is graded individually.
Special Information	The final grade for the DSE will be a combination of the group grade and the individual grade. Both the group grade and the individual grade will have to be 6.0 or higher in order to make the final grade valid. More detailed information on the grading including the grading criteria is made available before the start of the exercise. Grading takes place around the time of the mid term review and in the week after the exercise.
	Students will only be awarded a sufficient final grade for the project if: 1. The whole project has been completed and, 2. Both the group grade and the individual grade are 6.0 or higher.
Remarks	The final grade will be rounded to the nearest half grade. The final grade must be 6.0 or higher to pass the project. If a student fails the project (group grade and/or individual grade less than 6.0) he/she has to reregister to redo the project the following time the project is run. In that case grade will be registered as FAIL.
	Organization and assessment may differ due to COVID-19 measures. Changes to the organization and assessment method will be communicated via Brightspace.
Set-up	Student physical attendance and commitment is absolutely vital to the attainment of the best possible learning outcomes of the Bachelor thesis project, the DSE. The exercise requires a full time commitment (=40 hrs per week) of the student. It is recommended not to undertake any other study activity beside the exercise. Insufficient participation of an individual may lead to expulsion from the exercise.
	The following rules regarding absence hold: 1. Students are allowed to miss a maximum of 2 days for the entirety of the exercise. This is for unforeseen as well as foreseen circumstances. Additionally students may work one day from home. 2. Any missing time (either by being late or leaving at any time, irrespective its duration) during DSE will result in being registered as a missed day. 3. Absence, late arrival or early departure are recorded. Absence during obligatory key events of the DSE (e.g. Kick-off, Baseline, Mid-term and Final Review, DSE symposium, etc.) is not allowed under any condition. As soon as a student has been absent on more than two dates and has worked remotely for more than one day, no matter its cause or duration, he or she will be expelled from the exercise for the running period. Experience has shown that students spend all their time to the design project and have insufficient time left to make a serious preparation for an exam. Still, if a student wants to take an exam, the following conditions apply: The DSE project planning does not accommodate exams. The DSE absence regulations and project demands prevail; absence to prepare or take an exam has to be compliant with these regulations. Every absence has to be reported to the principal tutor. Every foreseen leave of absence has to be discussed first with the principal tutor during the first week of the exercise. Subsequently this has to be reported to the coordinating committee who will decide on the permission. During the exercise the rules of the house have to be obeyed. Violating these rules or showing a disrespectful attitude towards the academic or support staff or fellow student will lead to an official warning. When this happens a second time one will be expelled from the exercise. In case the DSE has to be executed online dedicated rules will be announced during the kick-off meeting prior to the start of the DSE. Students for whom the DSE is the last part of their BSc-curriculum can apply for receiving their BSc-diploma on the day of the concluding symposium. Eligibility will be judged based on progress up to and including Q3-exams (Spring DSE) or up to and including Q1-exams (Fall DSE). Detailed information will be provided during the DSE. Deadlines for making this possible will be adhered to strictly. The exercise takes ten full weeks and is offered twice a year. The first exercise runs in the second half of the first semester (November-January) and the second one in the second half of the second semester (April-July). Students are expected to put in a minimum of 40 hours of work a week into the exercise to a total of 400 hours. There is no room for other (study) activities (exams, minor courses, etc). The project introductions are given in week 1.6, 1.7 or 1.8 (Fall DSE) and week 3.6, 3.7 or 3.8 (Spring DSE). Attendance to this introduction is mandatory. At this introduction you will be given the opportunity to express your preference for specific design projects. The coordinating committee will try to match preferences with available projects. However, there are no guarantees that students will be assigned to the design project of their first preference. Normally students will be assigned to a design project in the top half of their list of preferred projects. The project ends with a symposium at which all the teams present their results.

AE3211	Aerospace Systems Engineering, Design & Production	6
Module Manager	Dr. F. Oliviero	
Contact Hours / Week x/x/x/x	Check AE3211-I and AE3211-II	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	The module AE3211 consists of two elements; a course on Systems Engineering and Aerospace Design (AE3211-I) and a course on Production of Aerospace Systems(AE3211-II). A detailed description of the contents and learning goals of these two courses can be found in the study guide, as well as information regarding practical matters (book, education method, etc.).	

AE3211-I	Systems Engineering & Aerospace Design	3
Responsible Instructor	Dr. F. Oliviero	
Instructor	Prof.dr. E.K.A. Gill	
Instructor	Dr. B.V.S. Jyoti	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Expected prior knowledge	First B.Sc. course year completed. AE2111 (both the modules AE2111-I and AE2111-II) completed.	
Parts	Lectures: 1. Systems Engineering for Aerospace 2. Systems Engineering Methods 3. Requirement and Functional Analysis I 4. Requirement and Functional Analysis II 5. Handling Design Iterations 6. Validation and Verification for Spacecraft I 7. Validation and Verification for Spacecraft II 8. Risk Management 9. Concurrent Engineering 10. Summary and Conclusions Study material: 1. Dynamical System Configuration 2. Technical and Project Management.	
Course Contents	- Characteristics of complex projects - Process of Systems Engineering - User needs and constraints - Setup of System Design Team - Requirement and functional analysis for aircraft/spacecraft mission - Categories of research and development work to match user needs - Design iterations - Risk management - Planning of verification efforts - Dynamical System Configuration - Technical and Project Management.	
Study Goals	The course shall provide the student with Systems Engineering material for engineering of complex aerospace products according to customer needs. In addition, it shall prepare the student for the Design Synthesis Exercise.	
Education Method	Lecture, Tutorial, Self-study.	
Literature and Study Materials	Lecture slides, material and example problems on Brightspace. Recommended literature 1. Applied Space Systems Engineering, Eds.: Larson, Kirkpatrick, Sellers, Thomas, Verma, McGraw-Hill, 2009 2. R. Hamann, M. van Tooren, Systems Engineering and Technical Management Techniques (I and II), Aerospace engineering faculty of TUDelft's publications 3. INCOSE Systems Engineering Handbook, International Council on Systems Engineering, version 2.0, July 2000 4. J.R. Wertz, W.J. Larson, Space mission analysis and design, Kluwer, Deventer, 1999, 3rd ed. ISBN stud. ed 0792359011. 5. B.S. Blanchard, W.J. Fabricky, Systems engineering and analysis, Prentice-Hall International, 1990, 2nd ed. ISBN 0138807582. 6. S. Jackson, Systems engineering for commercial aircraft, Ashgate publishing, 2002. Background literature 1. E. van Hinte, M. van Tooren, First Read This: Systems Engineering in Practice, 010 Publishers, Rotterdam.	
Assessment	Written exam, group tutorial. Both are mandatory. The participation to the group tutorial requires passing an individual preparation assignment. The final grade is computed as weighted average of the group tutorial grade and the individual exam grade. Detailed information on the assessment and grading policy is provided on Brightspace.	
Set-up	Ten lectures, self-study, Tutorial, Exam. Deliverables: week 7: Tutorial Preparation Report week 8: In-Class Tutorial Report	

AE3211-II	Production of Aerospace Systems	3
Responsible Instructor	Prof. C.A. Dransfeld	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	5	
Course Language	English	
Course Contents	I Part manufacture; the philosophy and physics of formative, subtractive and additive manufacturing processes, giving materials its shape and function. II Joining and integration methods of aerospace structures, such as bonding, riveting, bolting and welding. III Principles of sustainable manufacturing, determining and assessing the environmental cost of a manufacturing system. Recurring vs. non-recurring cost and automation principles. IV Cases studies (aviation/space/wind and emerging technologies). V Large scale manufacturing and assembly systems, lean manufacturing, principles of quality improvement.	
Study Goals	At the end of this course, the student will be able to: 1. Understand the principles of formative, subtractive and additive manufacturing and their impact on design freedom. 2. Theoretically apply combinations of formative, subtractive and additive manufacturing process and joining methods to realize complex structures. 3. Analyze manufacturing systems in respect to their environmental impact, considering the entire life cycle including end of life and evaluate scenarios with varying production volume and level of automation. 4. Apply principles of manufacturing system breakdown or decomposition to given case studies. 5. Remember principles of quality improvement in industrial large-scale manufacturing and assembly systems.	
Education Method	The lectures focus on the principles, tutorials help verify and expand the students level of understanding at greater detail.	
Literature and Study Materials	Lecture slides and tutorials. Some additional papers or articles, to expand on specific topics.	
Assessment	Written exam, closed books.	

AE3212	Aerospace Flight Dynamics, SVV including Flight Test	9
Module Manager	Dr.ir. A.C. in 't Veld	
Contact Hours / Week x/x/x/x	See AE3212-I and AE3212-II	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This module consists of two courses: AE3212-I Aerospace Flight Dynamics (including flight test) AE3212-II Simulation, Verification and Validation	
Study Goals	See individual courses	
Education Method	AE3212-I Lectures + test flight AE3212-II Lectures + group assignments	
Assessment	AE3212-I written exam AE3212-II written report	

AE3212-I	<i>Aerospace Flight Dynamics & Simulation, including Flight Test</i>	5
Responsible Instructor	Dr.ir. A.C. in 't Veld	
Instructor	C. Siemes	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Course Contents	Derivation and analysis of the stability and dynamic behavior of aerospace vehicles, taking into account the effects of different altitude/velocity combinations.	
Study Goals	At the end of this course, the student will be able to: 1. Determine the stability properties and characteristic motions of aerospace vehicles 2. Predict rotational motion of aerospace vehicles by means of simulation 3. Master reference frames + transformations 4. Utilize Euler angles 5. Derive and linearize the Equations of Motion 6. Determine static and dynamic stability properties 7. Separate symmetric/asymmetric E.O.M 8. Calculate and interpret Eigenvalues and Eigenmodes 9. Apply theory to different altitude/velocity regimes 10. Identify the effect of different vehicle configurations.	
Education Method	Lecturing and self study.	
Assessment	Combined written and online exam, on-site.	
Remarks	This course is the last in a series of three. It is preceded by Aerospace Design and Systems Engineering Elements I (AE1201) in BSc year 1 and followed by Aerospace Design and Systems Engineering Elements I (AE2101) in BSc year 2 The course is preceding the Design Synthesis (AE3200).	
Set-up	During the course students attend active lectures that cover relevant topics. Students are expected to have read the related course material in advance of class sessions; class sessions then consist of introduction of new theory, and exploration of the implications of the theory using real-life examples.	

AE3212-II	Simulation, Verification & Validation	4
Course Coordinator	Dr.ir. W. van der Wal	
Instructor	Dr.ir. A.C. in 't Veld	
Instructor	V. Yaghoubi Nasrabadi	
Contact Hours / Week x/x/x/x	0/0/12/0	
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	The following topics are addressed: - Creation of a numerical model for a structural analysis and a flight dynamics problem. - Verification of the numerical model against existing (analytical) models. - Assessment of the effect of assumptions on accuracy of results. - Validation of the simulation results with experiments.	
Study Goals	At the end of this course the student will be able to: 1. Plan the simulation, verification and validation of a complex aerospace problem. 2. Produce a complete numerical simulation to solve an aerospace problem, based on theory taught in courses of the BSc. 3. Verify the simulation with analytical models using a hierarchy of tests and to improve the simulation based on this verification. 4. Assess the influence of assumptions in the simulation. 5. Validate the simulation with data from experiments and improve the simulation based on this. 6. Organize group work and solve issues related to team work and group processes.	
Education Method	The course Simulation, Verification and Validation will be taught with a combination of teaching methods and will address topics from multiple domains. There will be two different group assignments, on the topics of Structural Analysis (SA) and Flight Dynamics (FD). The SA block will take 3 weeks, the FD block will take 3.5 weeks. For the SA assignment a verification and validation plan should be handed in at the end of week 1, which will be graded with a pass or fail. The FD block is the data processing part of the Flight Test (included in the course AE3212-I). Grades are given for reports for both assignments, to be handed in at the end of week 3 and in week 7 (deadlines to be announced).	
Prerequisites	The prerequisite is that all first and second-year projects must be completed in order to participate. To be able to participate in the group work and pass the course it is necessary to have a good knowledge of relevant parts of Structural Analysis (AE2135-I) and Numerical Analysis (AE2220), whether the courses are passed or not. Because the data processing of the 3rd-year Flight Test is also part of this course, following AE3212-I in parallel or having completed this course in previous years is highly recommended.	
Assessment	The assessment consists of 3 parts. The simulation plan produced in week 1 will be graded with a pass or fail, with a possibility to revise in order to receive a pass. The reports for the SA and FD assignments are graded. The grade for the FD report must be 5.8 or higher. In case the minimum grade is not achieved, a revision has to be submitted for the assignment report. The grades for this revision are scaled such that the maximum grade that can be achieved is 8.0, to be fair to groups that did not have to do a revision. A grade after revision for the SA part or for the FD part of 5.7 or lower means the student has to redo the entire course. The final grade is the weighted average of the grades for the SA report (weight 1/2) and FD report (weight 1/2).	
Enrolment / Application	Enrollment is mandatory. It will be announced to all BSc student when the enrollment opens and closes on Osiris.	
Set-up	The course requires 16 hours work per week, of which 12 hours are scheduled. Presence on-campus is mandatory for these hours with a maximum of two sessions that can be missed for the entire course. There are two lectures in the course, in the beginning of week 3.1 and in the beginning of week 3.4. In the first lecture students simulation, verification, and validation will be introduced. After that there is a recap of SA knowledge and introduction of the first assignment. The FD block is the data processing part of the Flight Test (included in the course AE3212-I). Groups of 8 students are formed by the course coordinators before the first lecture. At the day of the first lecture all students have to confirm that they participate in the group work or they risk not being allowed to continue in the course. This is to avoid as much as possible having groups of less than 8 students. Incomplete groups with 6 students or less might be divided over other groups. During the scheduled course hours a project table is reserved for the group and TA's are available to answer questions. At the end of the first lecture, the assignment is made available. In the remainder of the first week, the group will work on a verification and validation plan. This plan will be graded with a pass or fail and will serve as the basis for the project work. Additional help is available in the form of a monitored forum on Brightspace. Week 2 and 3 consist of unsupervised project work which is concluded by a written report that is graded as explained in 'assessment'. Before week 4 new groups are made for the Flight Dynamics (FD) assignment. These groups consist of 6 students and they are the same as the groups for the flight test. The lecture in week 4 consists of FD knowledge for the flight test, practical knowledge for the flight tests, and an introduction of the assignment. Students that want to participate in the flight test must attend this lecture. Unfortunately no exceptions can be made because this lecture is also the briefing for the flight test. After the lecture group work starts. The groups will work on a verification and validation plan. This plan must be handed in at the end of week 3.4. It is not graded, but you will receive feedback. The deadline for the FD report is in week 7 of the course (exact deadline to be announced). The report is graded as explained in 'assessment'. COVID-19: If required due to COVID-19 measures, lectures and help sessions with TA's will take place online. In all other cases, lectures and project sessions are on campus, and presence is mandatory.	

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